

AWS Academy Cloud Foundations

Module 8: Databases



Topics

- Amazon Relational Database Service (Amazon RDS)
- Amazon DynamoDB
- Amazon Redshift
- Amazon Aurora

Demos

- Amazon RDS console
- Amazon DynamoDB console

Lab

- Lab 5: Build Your DB Server and Interact with Your DB Using an App

Activity

- Database case studies



Knowledge check

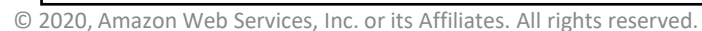
Module objectives

After completing this module, you should be able to:

- Explain Amazon Relational Database Service (Amazon RDS)
- Identify the functionality in Amazon RDS
- Explain Amazon DynamoDB
- Identify the functionality in Amazon DynamoDB
- Explain Amazon Redshift
- Explain Amazon Aurora
- Perform tasks in an RDS database, such as launching, configuring, and interacting

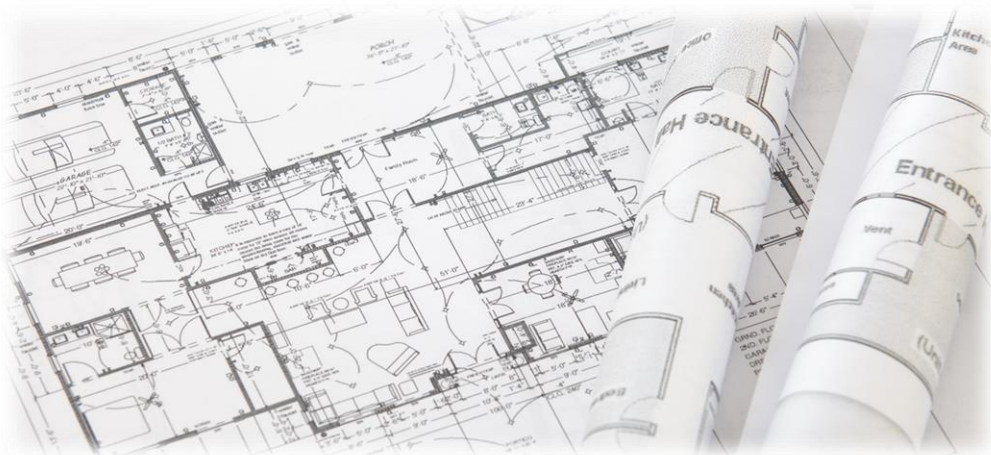
Module 8: Databases

Section 0: Architectural need and Database layer considerations



Café business requirement

The café needs a database solution that is easier to maintain, and that provides essential features such as durability, scalability, and high performance.

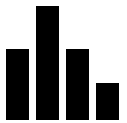




Scalability



Total storage requirements



Object size and type



Durability

How much throughput is needed?

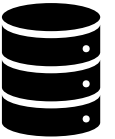
Will the chosen solution be able to scale up later, if needed?



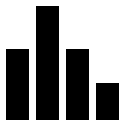
Database considerations: Storage requirements



Scalability



Total storage requirements



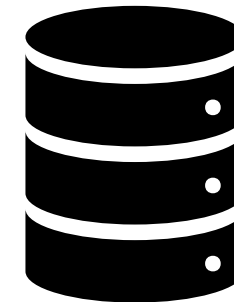
Object size and type



Durability

How large does the database need to be?

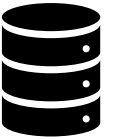
Will it need to store GB, TB, or petabytes of data?



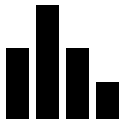
Database considerations: Object size and type



Scalability



Total storage requirements

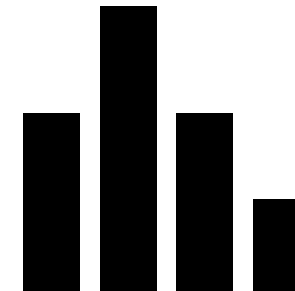


Object size and type



Durability

**Do you need to store simple data structures,
large data objects, or both?**



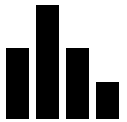
Database considerations: Durability



Scalability



Total storage requirements



Object size and type



Durability

What level of **data durability**, **data availability**, and **recoverability** is required?

Do regulatory obligations apply?



Now that you reviewed key considerations, consider the two categories of database options available:

Relational

Traditional examples:

Microsoft SQL Server
Oracle Database
MySQL

Non-Relational

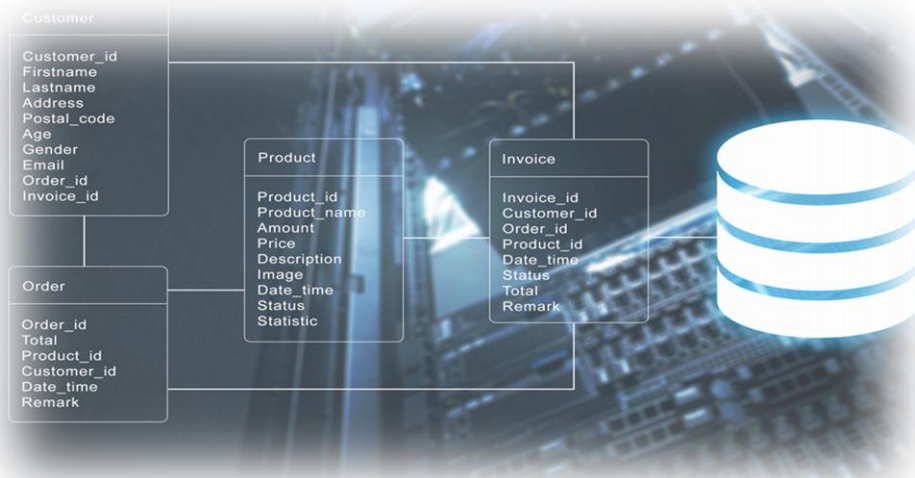
Traditional examples:

MongoDB
Cassandra
Redis

Relational database type

Benefits:

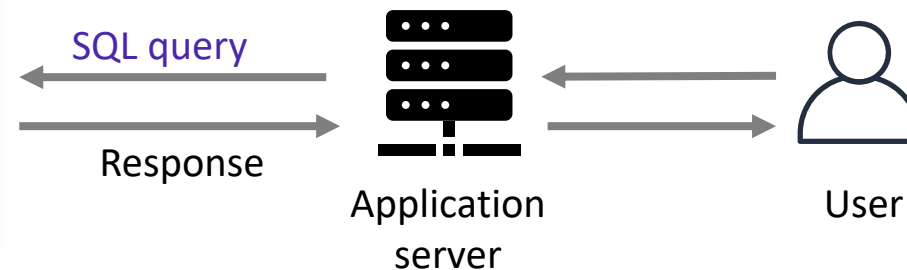
- Ease of use
- Data integrity
- Reduced data storage
- Common language (structured query language, or SQL)



Relational database management system (RDBMS)

Relational is ideal when you:

- Need strict schema rules, ACID compliance, and data quality enforcement
- Do not need extreme read/write capacity
- Do not need extreme performance
 - An RDBMS can be the best, lowest-effort solution



Non-relational database type

Benefits

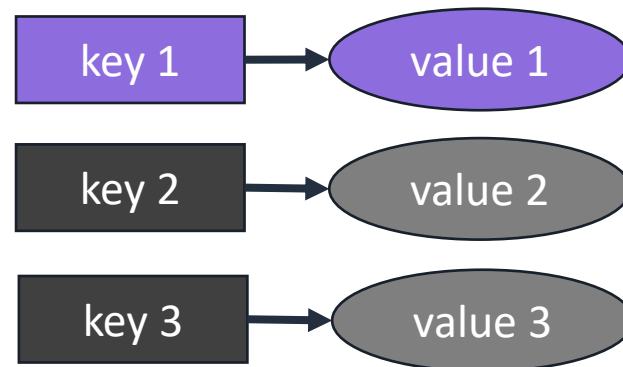
- Flexibility
- Scalability
- High performance
- Highly functional APIs

Non-relational is ideal when:

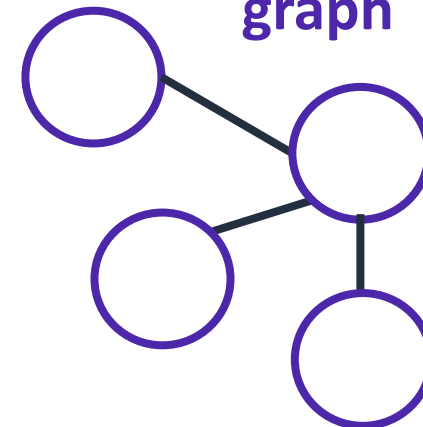
- Database must scale **horizontally** to handle massive data volume
- Data does not lend itself well to traditional schemas
- Read/write rates exceed what can be economically supported through traditional RDBMS

Example models

key-value



graph



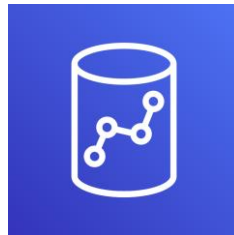
Amazon database options

More database options exist—these options are common examples

Relational databases



Amazon
RDS

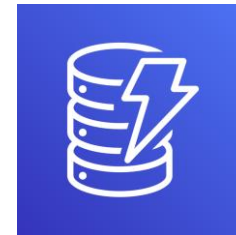


Amazon
Redshift



Amazon
Aurora

Non-relational databases



Amazon
DynamoDB



Amazon
ElastiCache



Amazon
Neptune

Focus in this module

Section 0 key takeaways



- When you choose a database, consider **scalability**, **storage requirements**, the **type and size of objects** to be stored, and **durability requirements**
- **Relational databases** have strict schema rules, provide data integrity, and support SQL
- **Non-relational databases** scale horizontally, provide higher scalability and flexibility, and work well for semistructured and unstructured data

Module 8: Databases

Section 1: Amazon Relational Database Service



Amazon Relational Database Service (Amazon RDS)

Unmanaged versus managed services

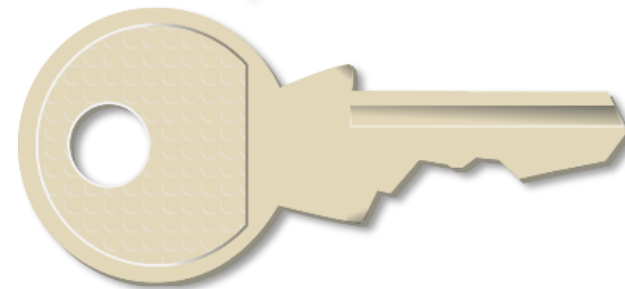
Unmanaged:

Scaling, fault tolerance, and availability are managed by you.



Managed:

Scaling, fault tolerance, and availability are typically built in to the service.

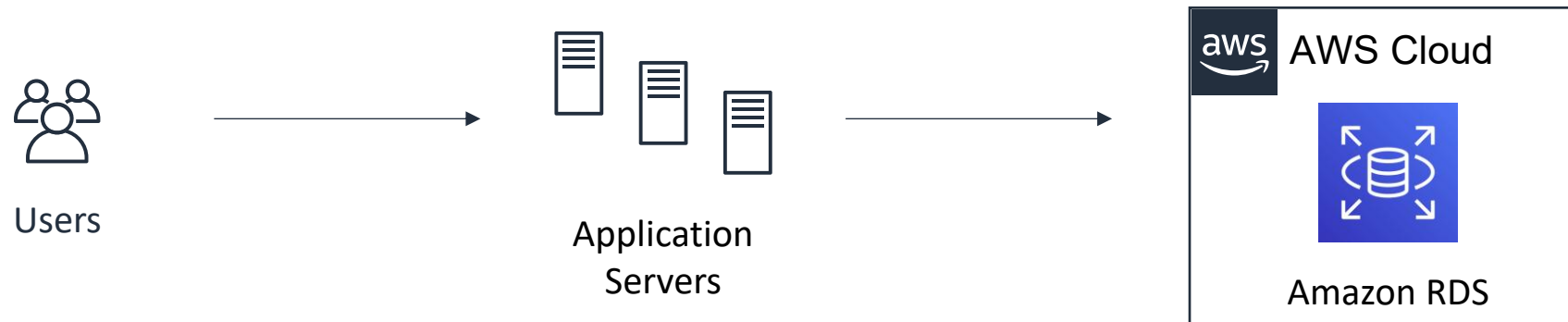


Challenges of relational databases

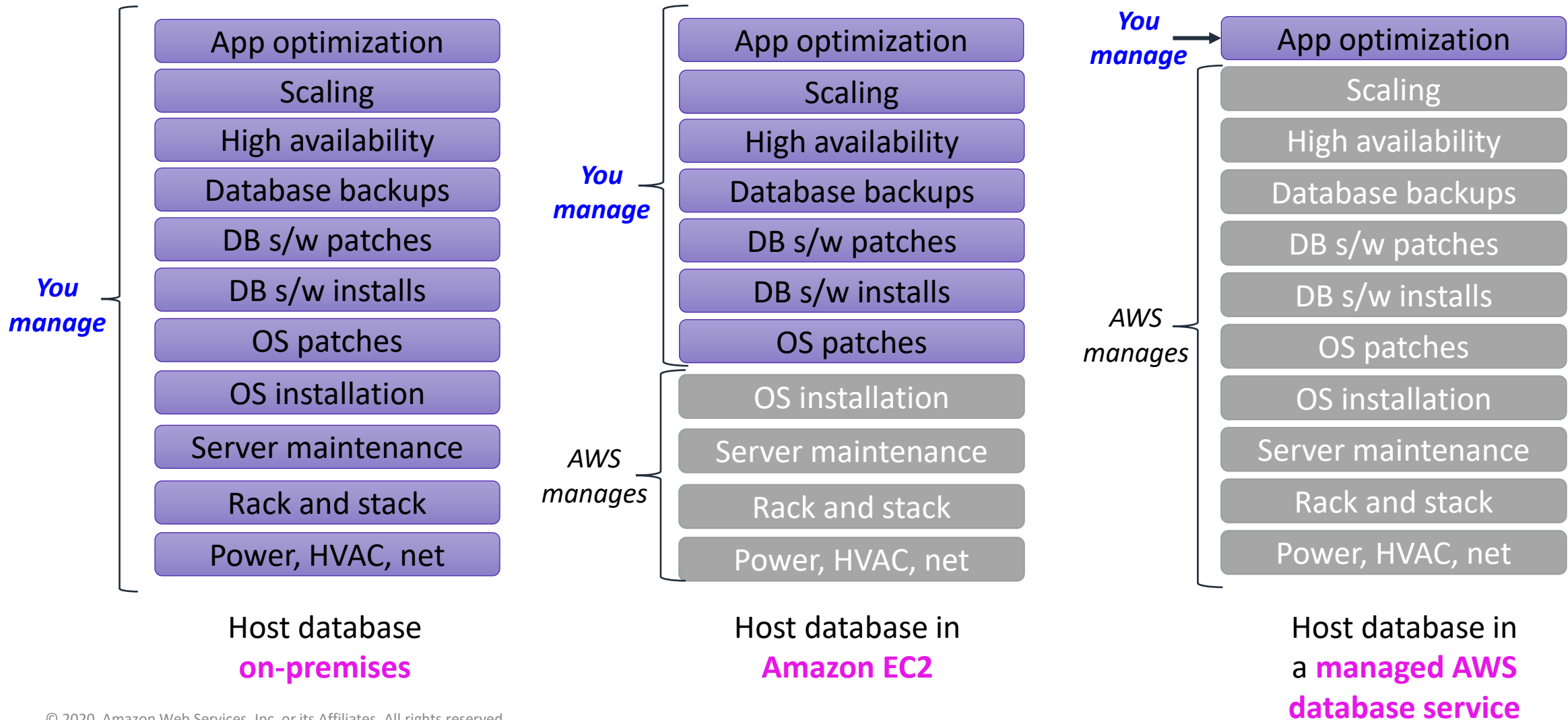
- Server maintenance and energy footprint
- Software installation and patches
- Database backups and high availability
- Limits on scalability
- Data security
- Operating system (OS) installation and patches



Managed service that sets up and operates a relational database in the cloud.



Advantage of managed AWS database services



Managed services responsibilities

You manage:

- Application optimization



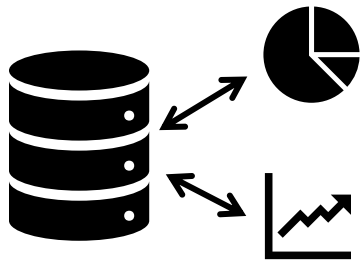
AWS manages:

- OS installation and patches
- Database software installation and patches
- Database backups
- High availability
- Scaling
- Power and racking and stacking servers
- Server maintenance



Amazon RDS

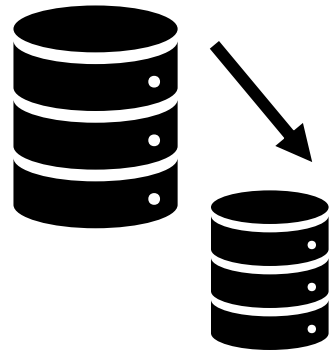
Amazon RDS characteristics



Access pattern

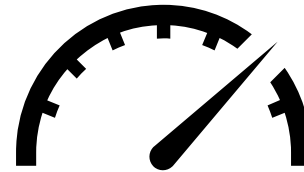
Transactional

Light analytics



Data size

Low-TB range



Performance

Mid to high throughput

Low latency



Business use cases

Transactional

OLAP

Amazon RDS: Uses and database types



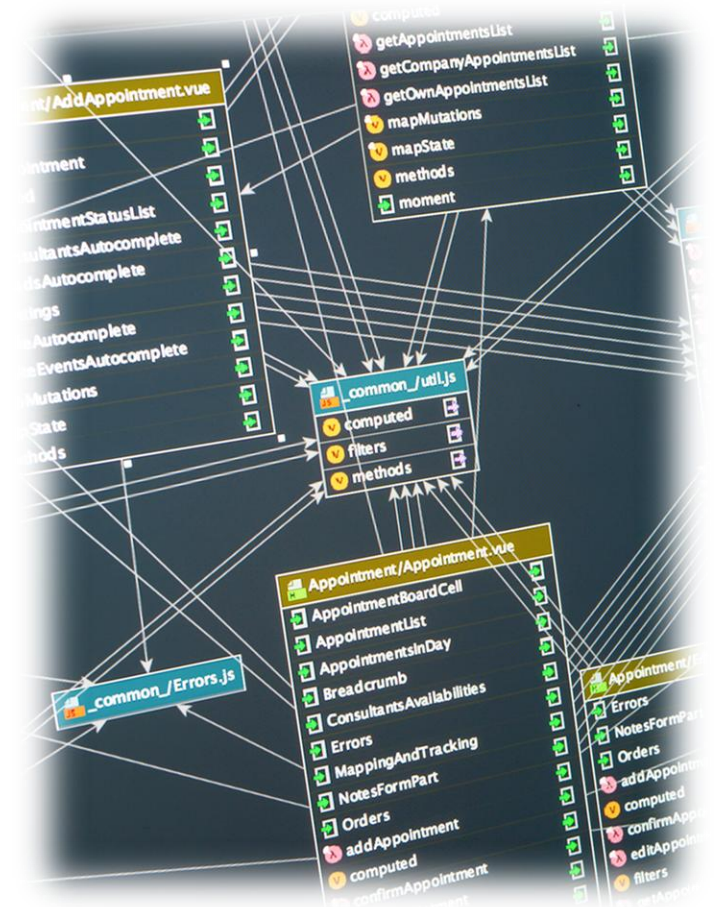
Amazon
RDS

Works well for applications that:

- Have more complex data
- Need to combine and join datasets
- Need enforced syntax rules

Six database types supported:

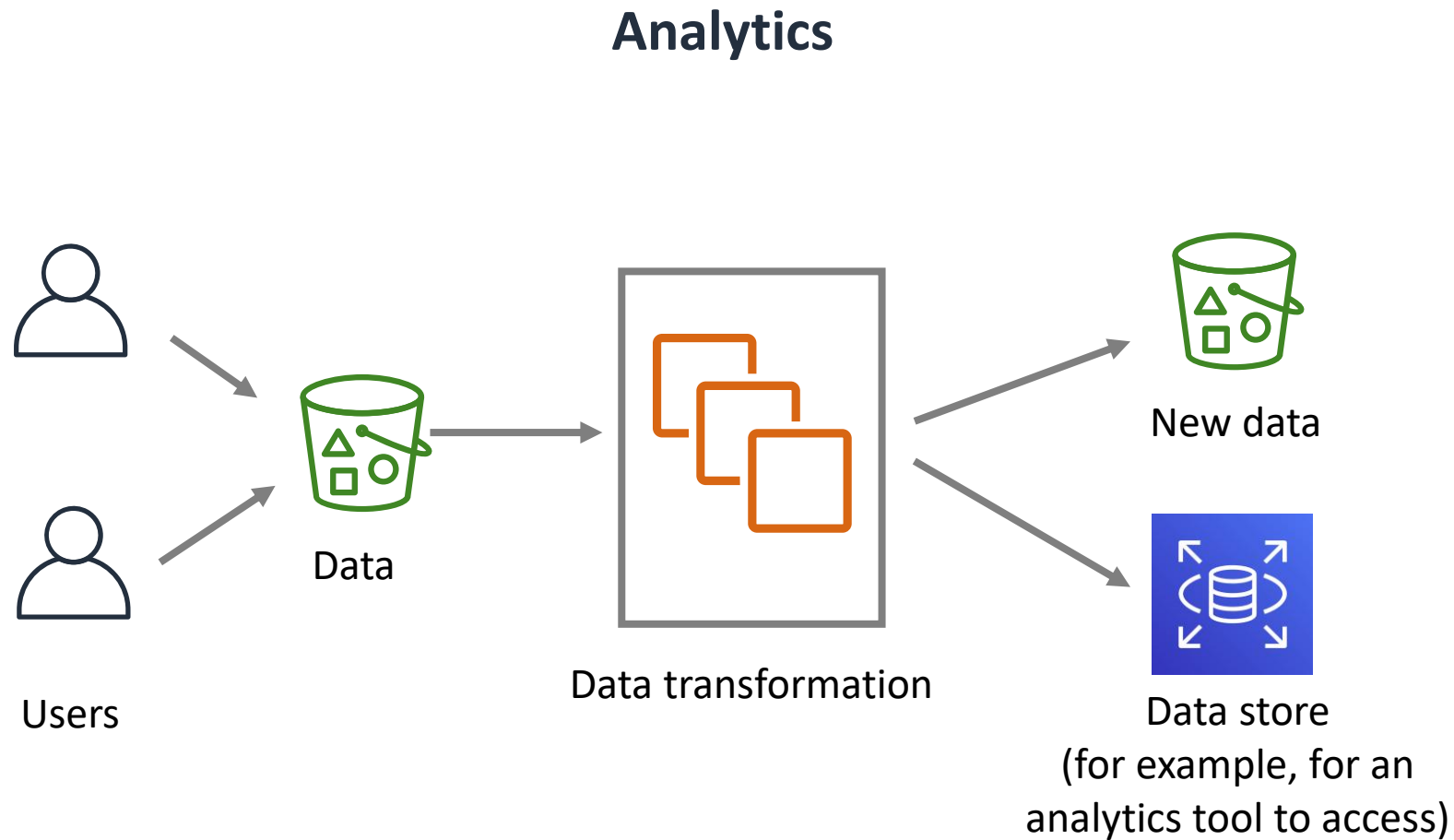
- Microsoft SQL Server
- PostgreSQL
- Oracle
- Aurora
- MySQL
- MariaDB



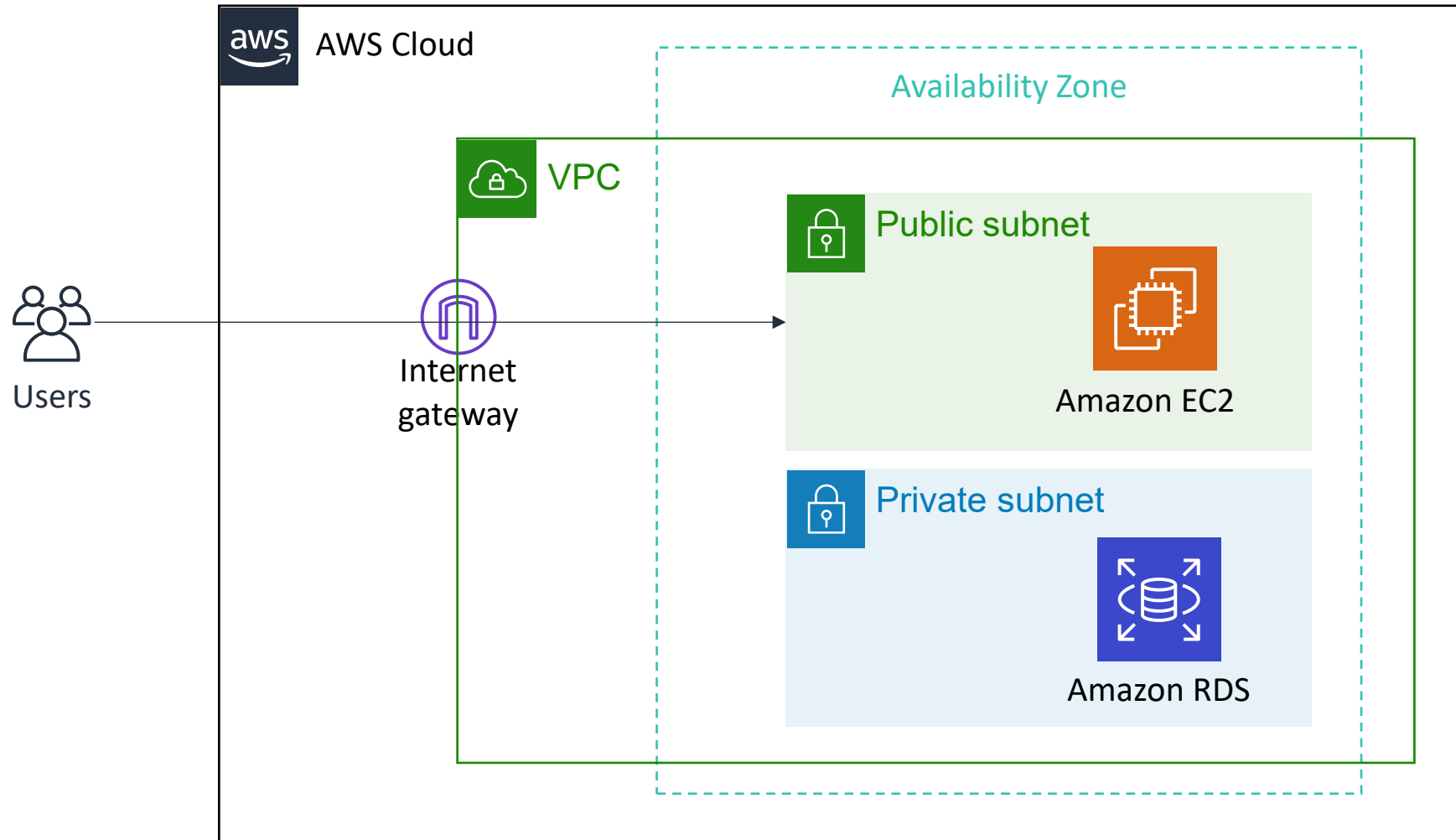
Database instance sizing

	T family	M family	R family
Type	Burstable instances	General-purpose instances	Memory-optimized instances
Sizing	1 vCPU/1 GB RAM to 8 vCPU 32 GB RAM	2 vCPU/8 GB RAM to 96 vCPU 384 GB RAM	2 vCPU/16 GB RAM to 96 vCPU 768 GB RAM
Networking	Moderate performance	High performance	High performance
Ideal Workload	Smaller or variable	CPU-intensive	Query-intensive, high connection counts
Highlights	T3 can burst above baseline for extra charge	M5 offers up to 96 vCPU	R5 offers up to 96 vCPU 768 GiB RAM

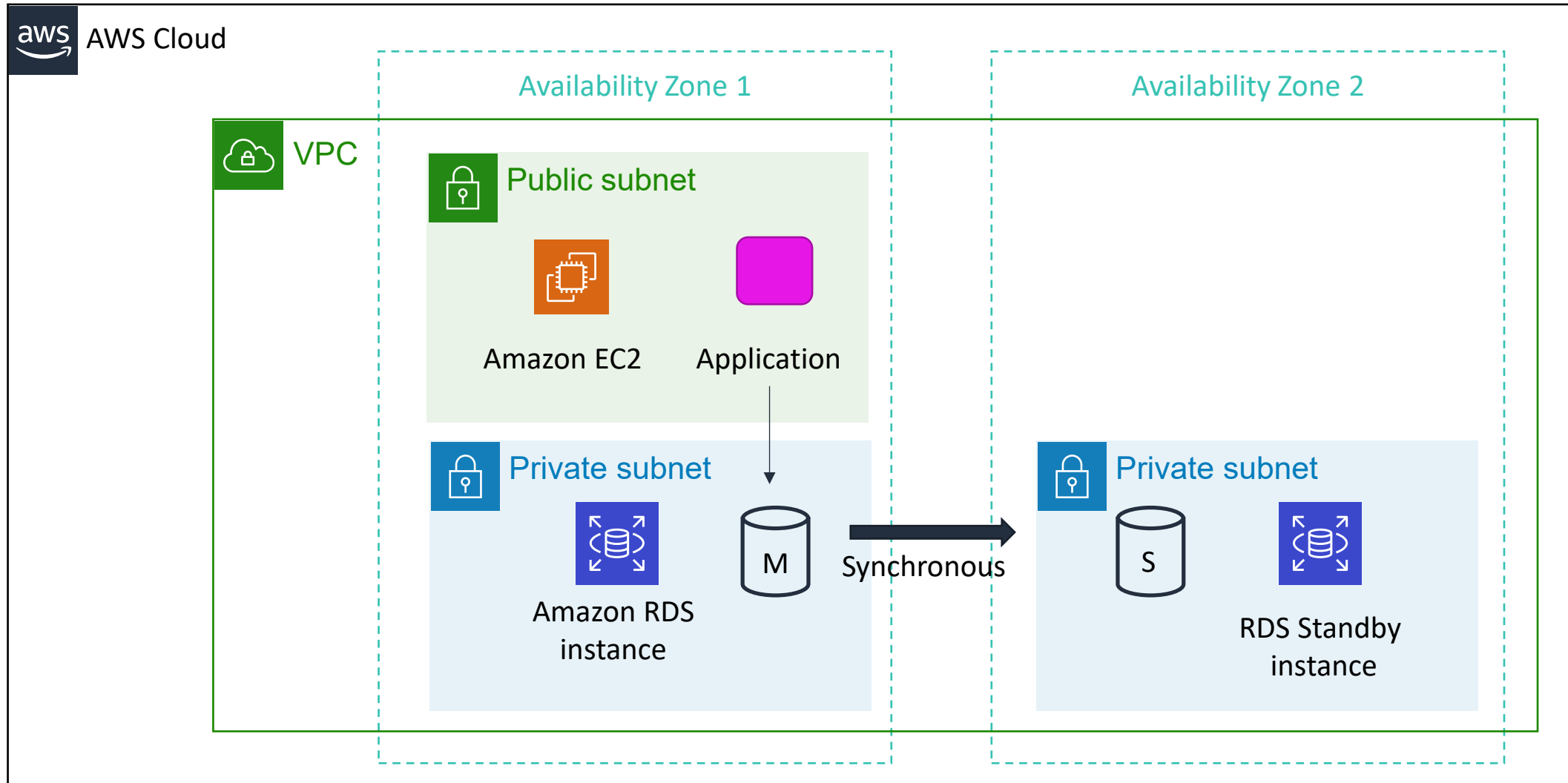
Amazon RDS: Example use case



Amazon RDS in a virtual private cloud (VPC)

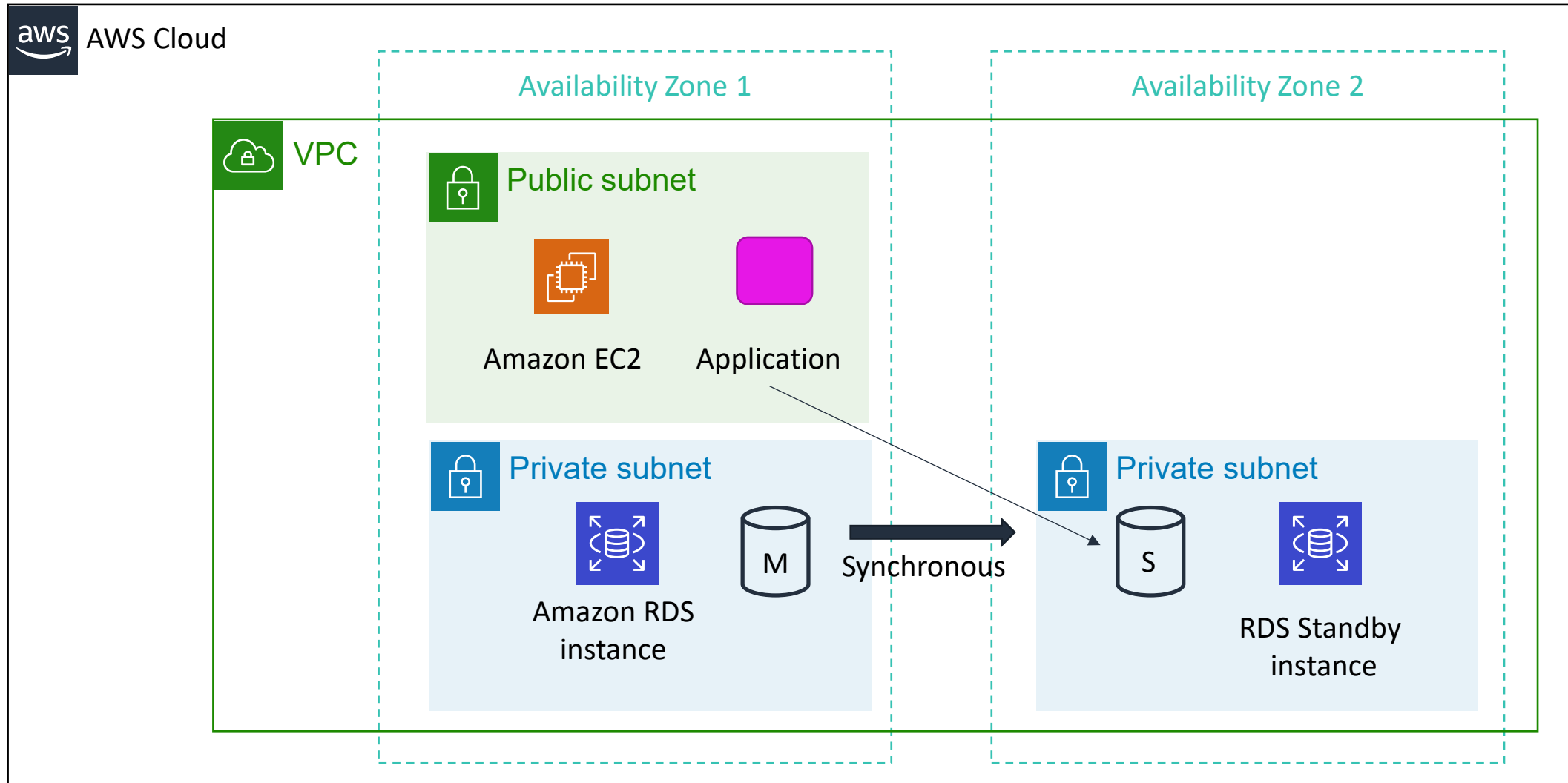


High availability with Multi-AZ deployment



High availability with Multi-AZ deployment

2



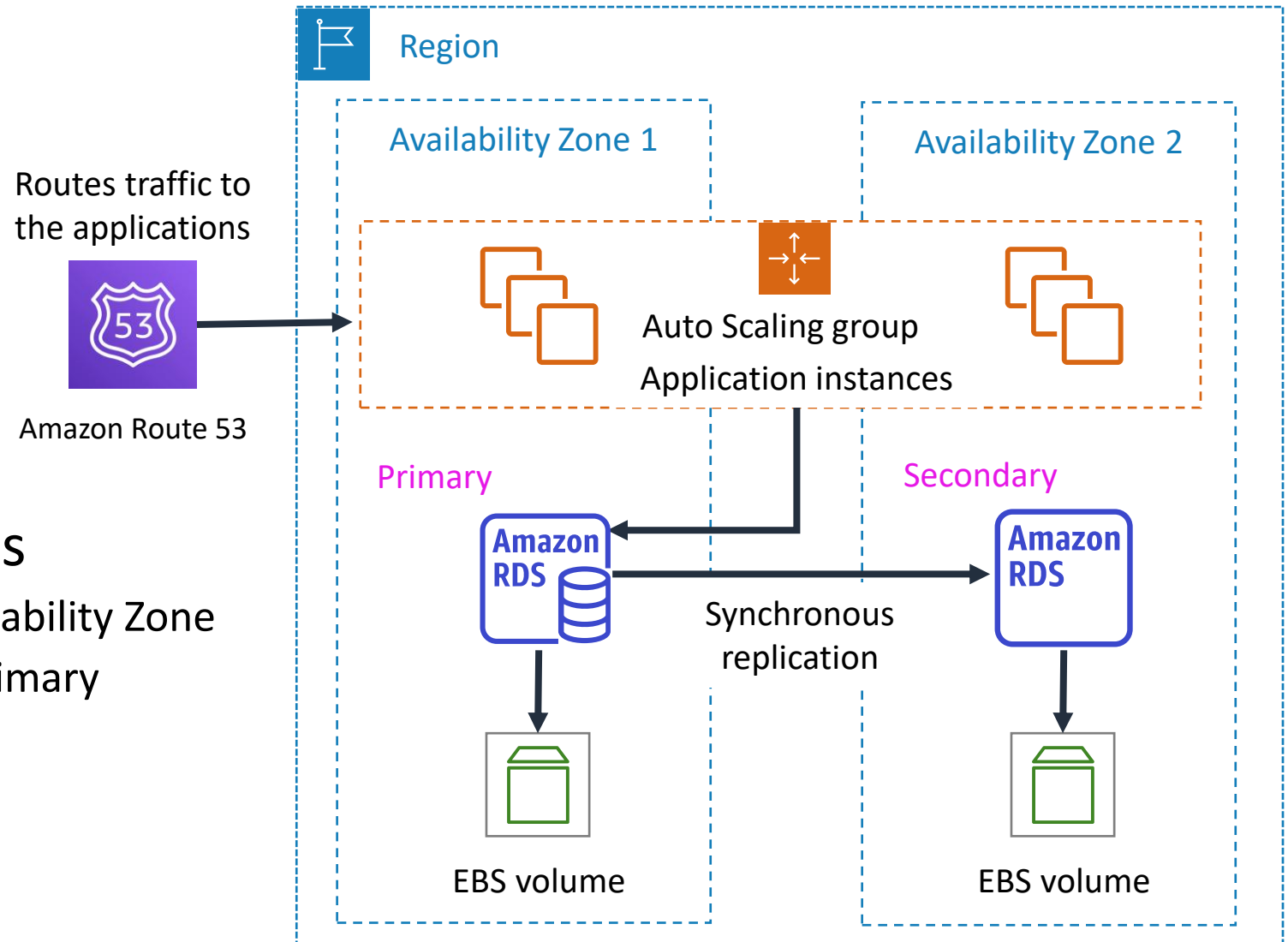
Multi-AZ deployment for high availability

Benefits

- Enhanced durability
- Increased availability
- Fail over to standby occurs automatically

Automated failover conditions

- Loss of availability in primary Availability Zone
- Loss of network connectivity to primary
- Compute unit failure on primary
- Storage failure on primary



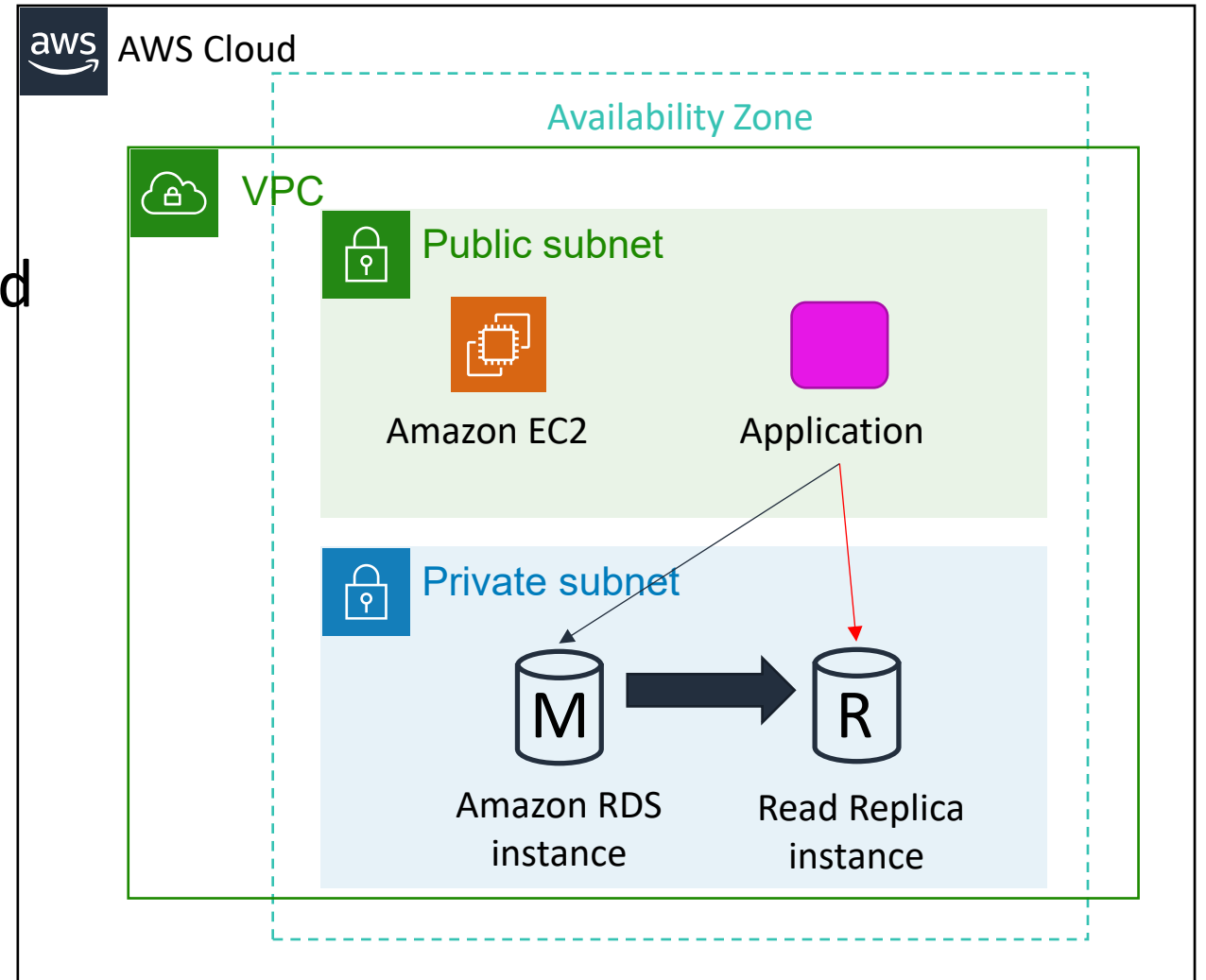
Amazon RDS read replicas

Features

- Offers asynchronous replication
- Can be promoted to master if needed

Functionality

- Use for read-heavy database workloads
- Offload read queries



Read replicas for performance

Benefits

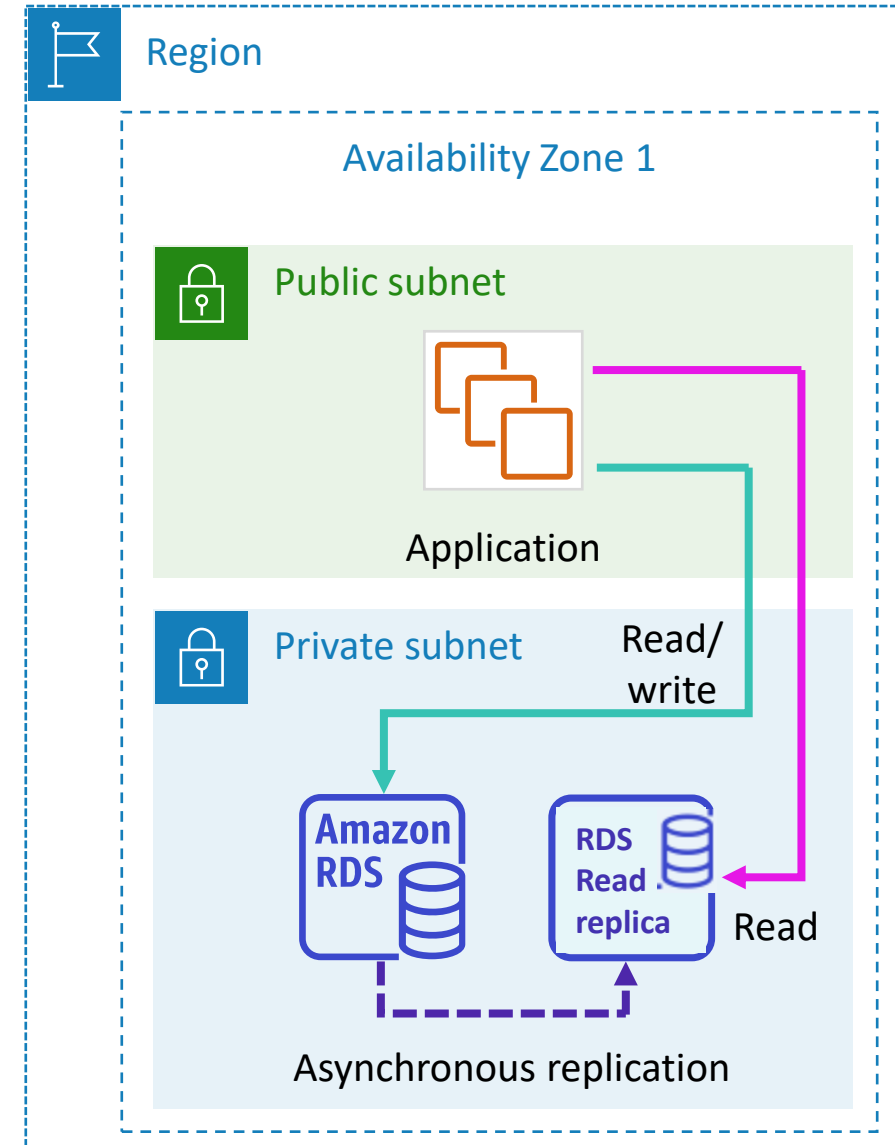
- Enhanced performance
- Increased availability
- Designed for security

Supported by

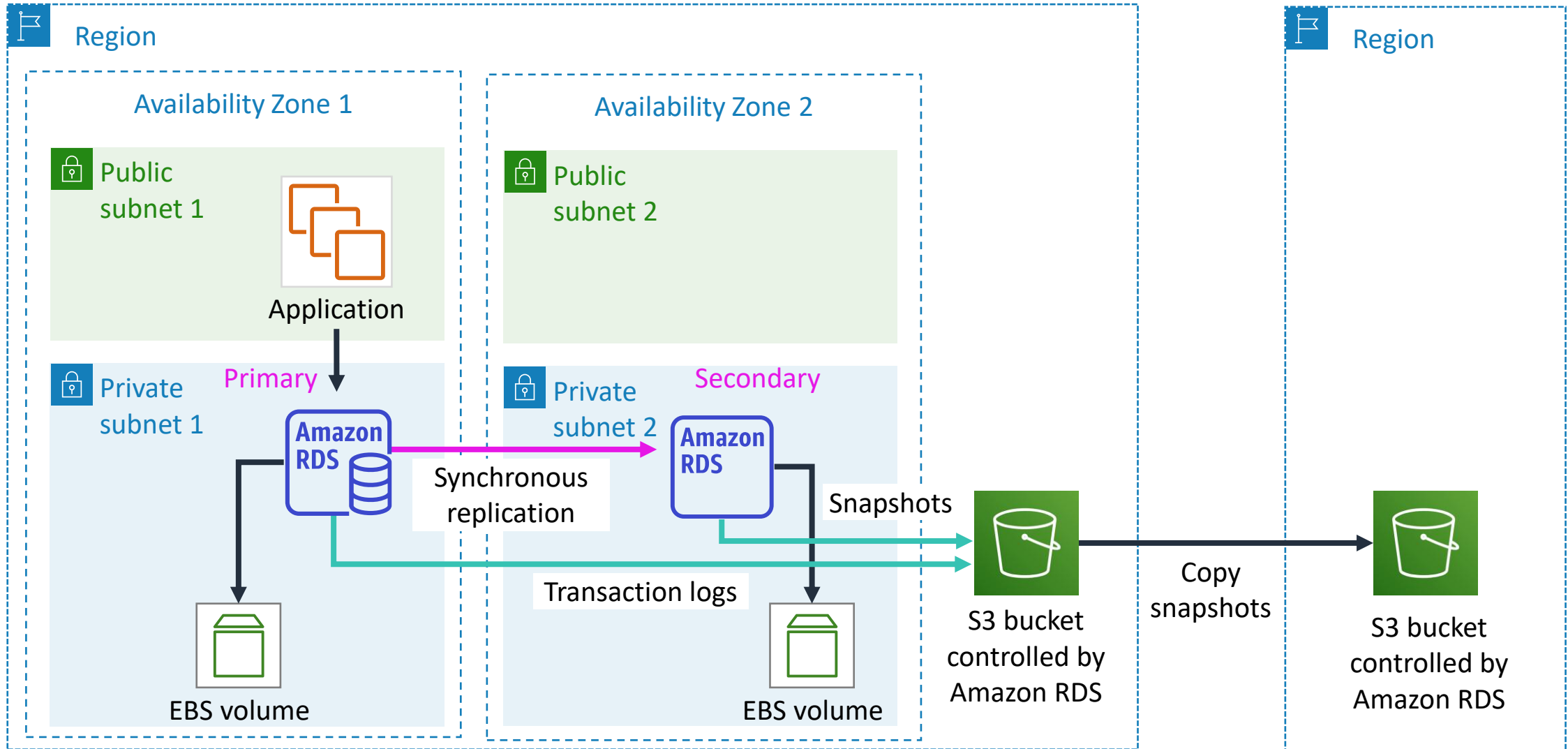
- MySQL
- MariaDB
- PostgreSQL
- Oracle

Limits

- Five read replicas per primary
- For strict read-after-write consistency, read from the primary



Amazon RDS backup solution



Web and mobile applications	✓ High throughput ✓ Massive storage scalability ✓ High availability
Ecommerce applications	✓ Low-cost database ✓ Data security ✓ Fully managed solution
Mobile and online games	✓ Rapidly grow capacity ✓ Automatic scaling ✓ Database monitoring

When to Use Amazon RDS

Use Amazon RDS when your application requires:

- Complex transactions or complex queries
- A medium to high query or write rate – Up to 30,000 IOPS (15,000 reads + 15,000 writes)
- No more than a single worker node or shard
- High durability

Do not use Amazon RDS when your application requires:

- Massive read/write rates (for example, 150,000 write/second)
- Sharding due to high data size or throughput demands
- Simple GET or PUT requests and queries that a NoSQL database can handle
- Relational database management system (RDBMS) customization

Amazon RDS: Clock-hour billing and database characteristics



Clock-hour billing –

- Resources incur charges when running

Database characteristics –

- Physical capacity of database:
 - Engine
 - Size
 - Memory class

Amazon RDS: DB purchase type and multiple DB instances



DB purchase type –

- On-Demand Instances
 - Compute capacity by the hour
- Reserved Instances
 - Low, one-time, upfront payment for database instances that are reserved with a 1-year or 3-year term

Number of DB instances –

- Provision multiple DB instances to handle peak loads

Provisioned storage –

- No charge
 - Backup storage of up to 100 percent of database storage for an active database
- Charge (*GB/month*)
 - Backup storage for terminated DB instances

Additional storage –

- Charge (*GB/month*)
 - Backup storage in addition to provisioned storage

Amazon RDS: Deployment type and data transfer



Requests –

- The number of input and output requests that are made to the database

Deployment type—Storage and I/O charges vary, depending on whether you deploy to –

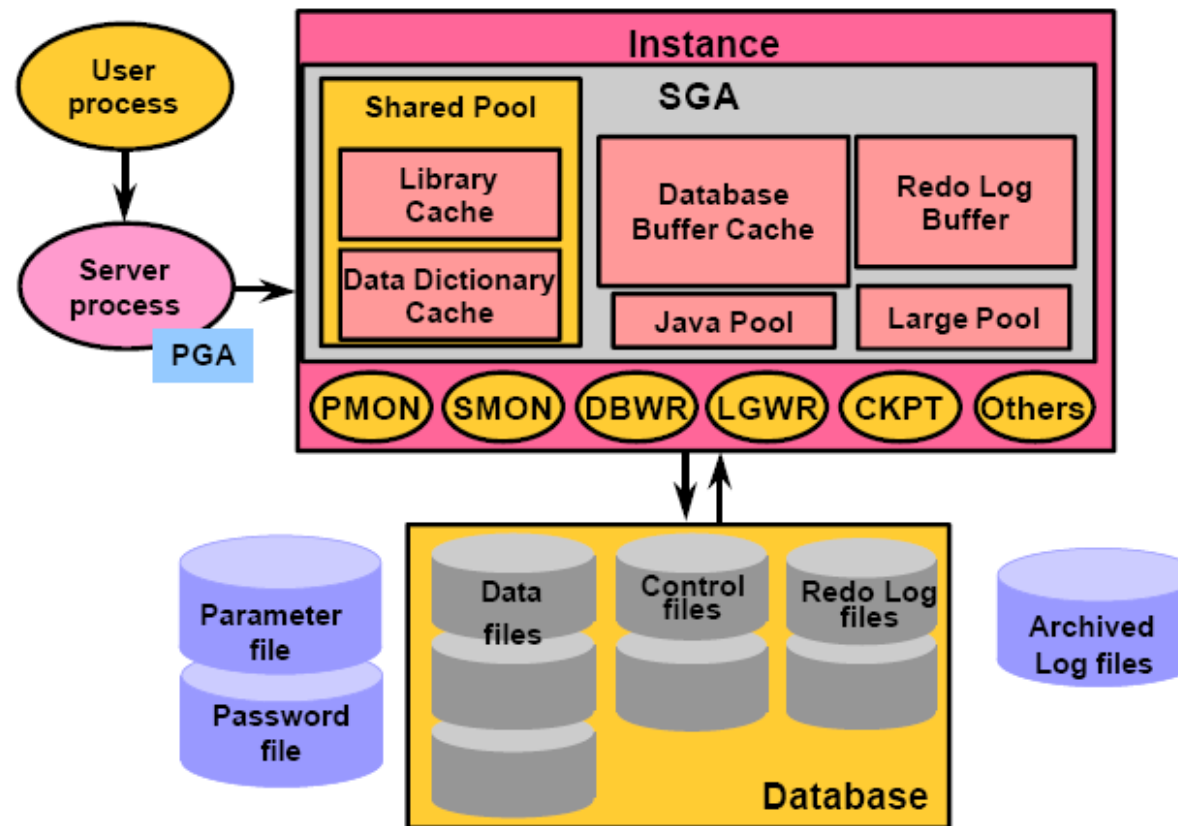
- Single Availability Zone
- Multiple Availability Zones

Data transfer –

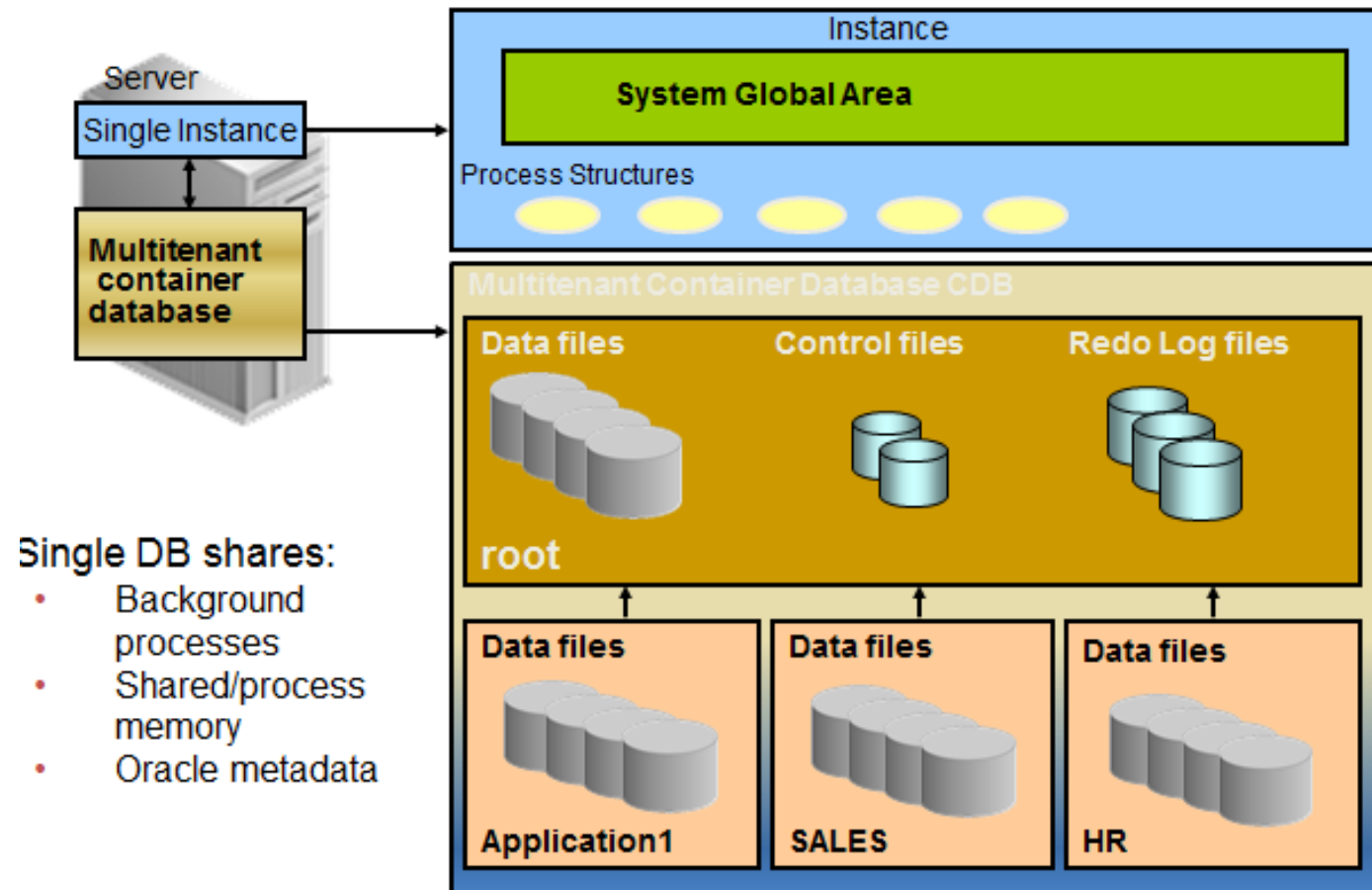
- No charge for inbound data transfer
- Tiered charges for outbound data transfer

Non-MultiTenant Architecture Diagram

Overview of Primary Components

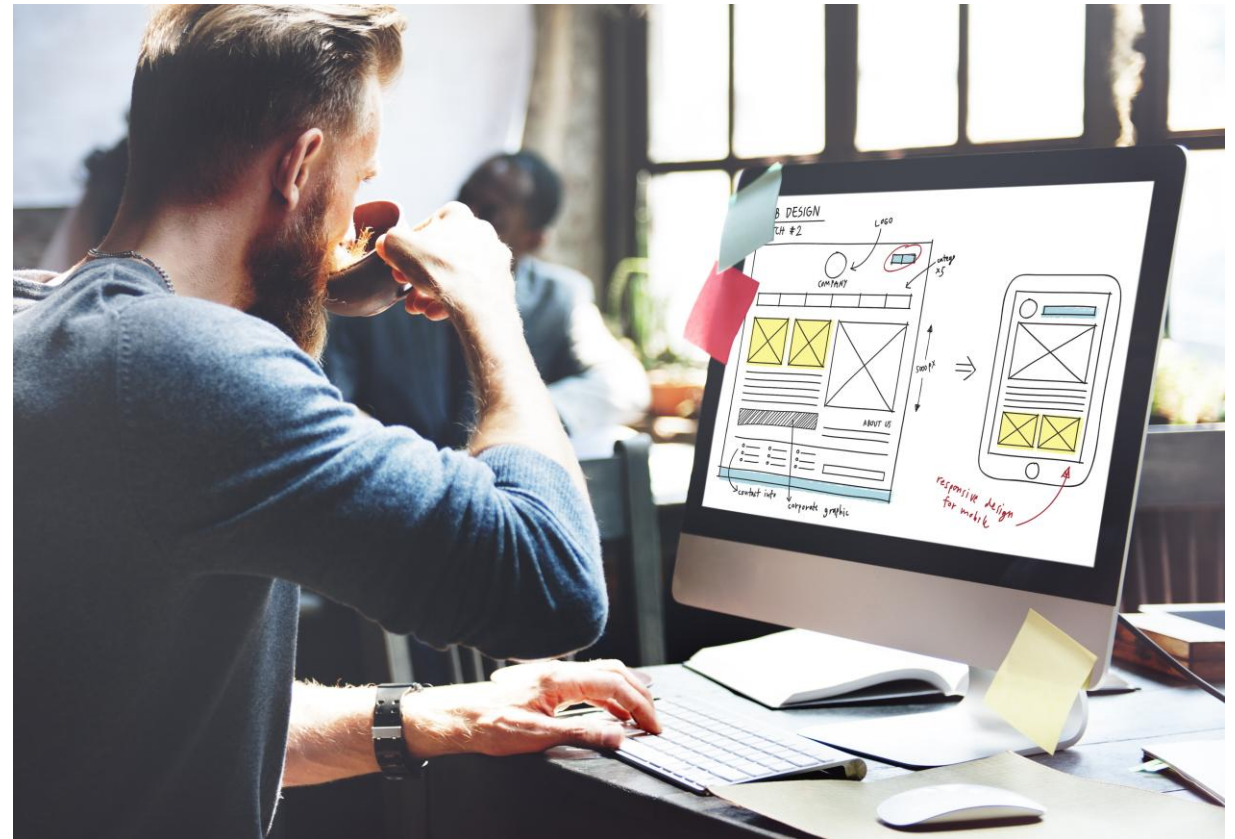


MultiTenant Architecture Diagram



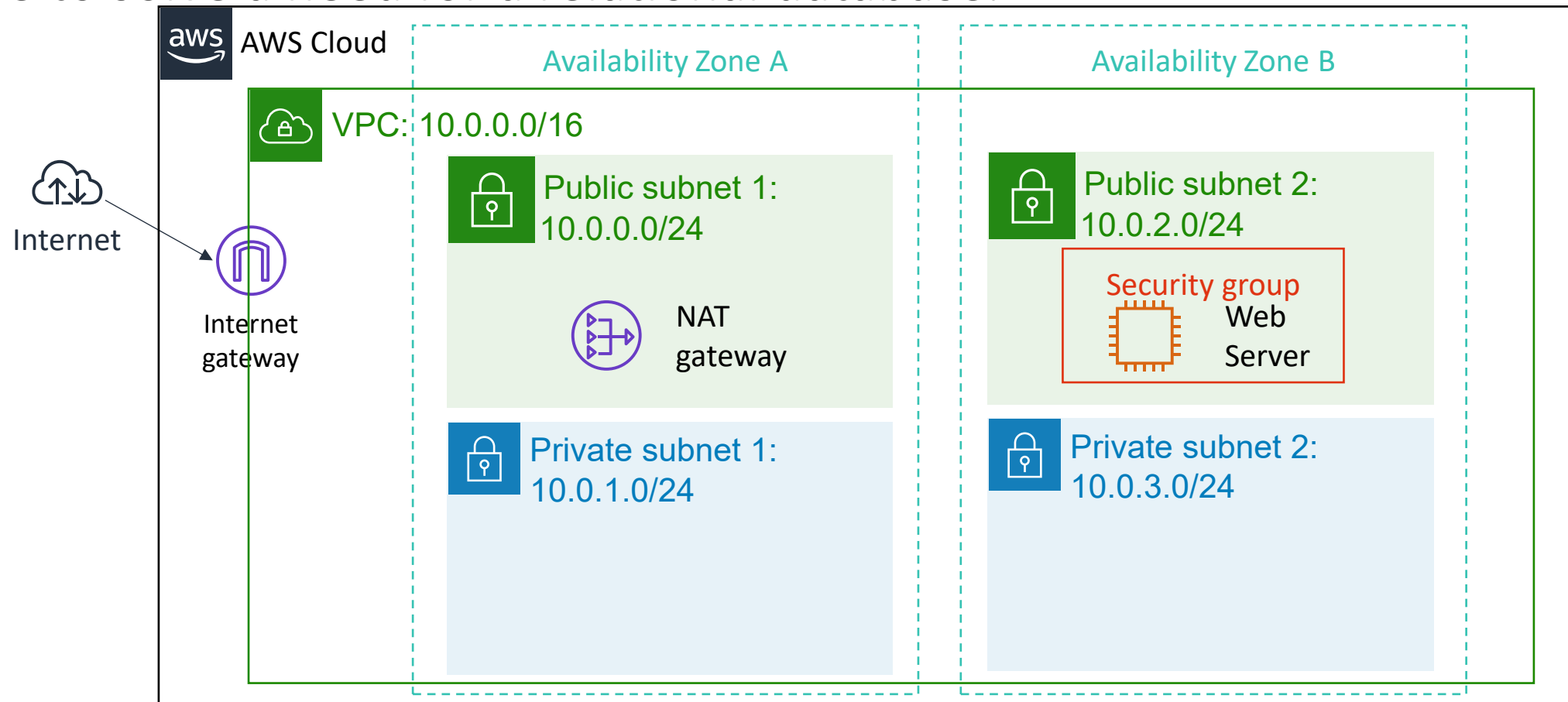
<https://docs.oracle.com/database/122/CNCPT/overview-of-the-multitenant-architecture.htm#CNCPT89257>

Build Your DB Server and Interact with Your DB Using an App



Lab 5: Scenario

This lab is designed to show you how to use an AWS managed database instance to solve a need for a relational database.



Security group

Create a **VPC security group**.



Private subnet

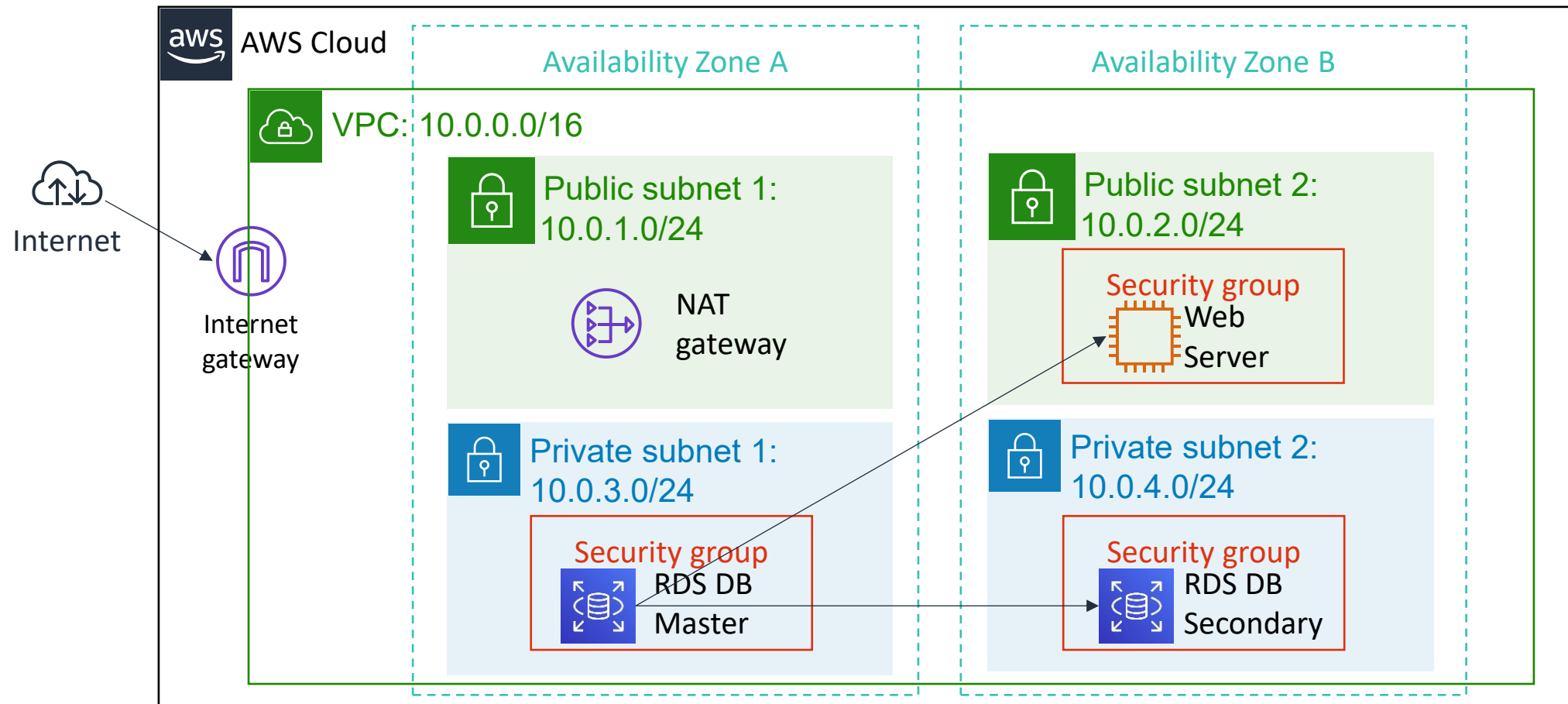
Create a **DB subnet group**.



Amazon RDS

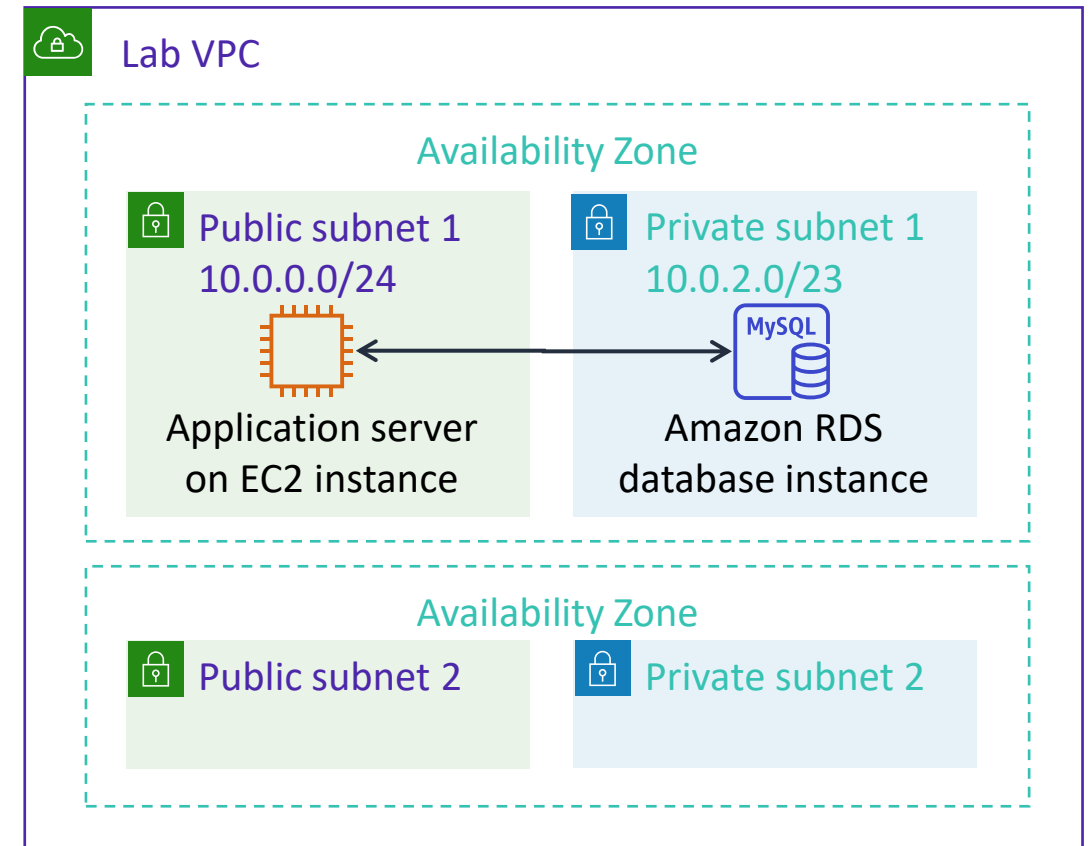
Create an **Amazon RDS DB** instance and interact with your database.

Lab 5: Final product



Guided lab: Tasks (ACA Module 5)

1. Creating an Amazon RDS database
2. Configuring web application communication with a database instance





Begin Lab 5: Build your DB server and interact with your DB using an application

Lab debrief: key takeaways



Recorded demo: Amazon RDS



Set up demo

Amazon Relational Database Service (RDS)

Section 1 key takeaways



- With Amazon RDS, you can set up, operate, and scale relational databases in the cloud.
- Features –
 - Managed service
 - Accessible via the console, AWS Command Line Interface (AWS CLI), or application programming interface (API) calls
 - Scalable (compute and storage)
 - Automated redundancy and backup are available
 - Supported database engines:
 - Amazon Aurora, PostgreSQL, MySQL, MariaDB, Oracle, Microsoft SQL Server

Module 8: Databases

Section 2: Amazon DynamoDB

Relational versus non-relational databases

	Relational (SQL)	Non-Relational												
Data Storage	Rows and columns	Key-value, document, graph												
Schemas	Fixed	Dynamic												
Querying	Uses SQL	Focuses on collection of documents												
Scalability	Vertical	Horizontal												
Example	<table><tr><th>ISBN</th><th>Title</th><th>Author</th><th>Format</th></tr><tr><td>3111111223439</td><td>Withering Depths</td><td>Jackson, Mateo</td><td>Paperback</td></tr><tr><td>3122222223439</td><td>Wily Willy</td><td>Wang, Xiulan</td><td>Ebook</td></tr></table>	ISBN	Title	Author	Format	3111111223439	Withering Depths	Jackson, Mateo	Paperback	3122222223439	Wily Willy	Wang, Xiulan	Ebook	<pre>{ ISBN: 3111111223439, Title: "Withering Depths", Author: "Jackson, Mateo", Format: "Paperback" }</pre>
ISBN	Title	Author	Format											
3111111223439	Withering Depths	Jackson, Mateo	Paperback											
3122222223439	Wily Willy	Wang, Xiulan	Ebook											

What is Amazon DynamoDB?

Fast and flexible NoSQL database service for any scale



Amazon DynamoDB

- NoSQL database tables
- Virtually unlimited storage
- Items can have differing attributes
- Low-latency queries
- Scalable read/write throughput

What is Amazon DynamoDB?

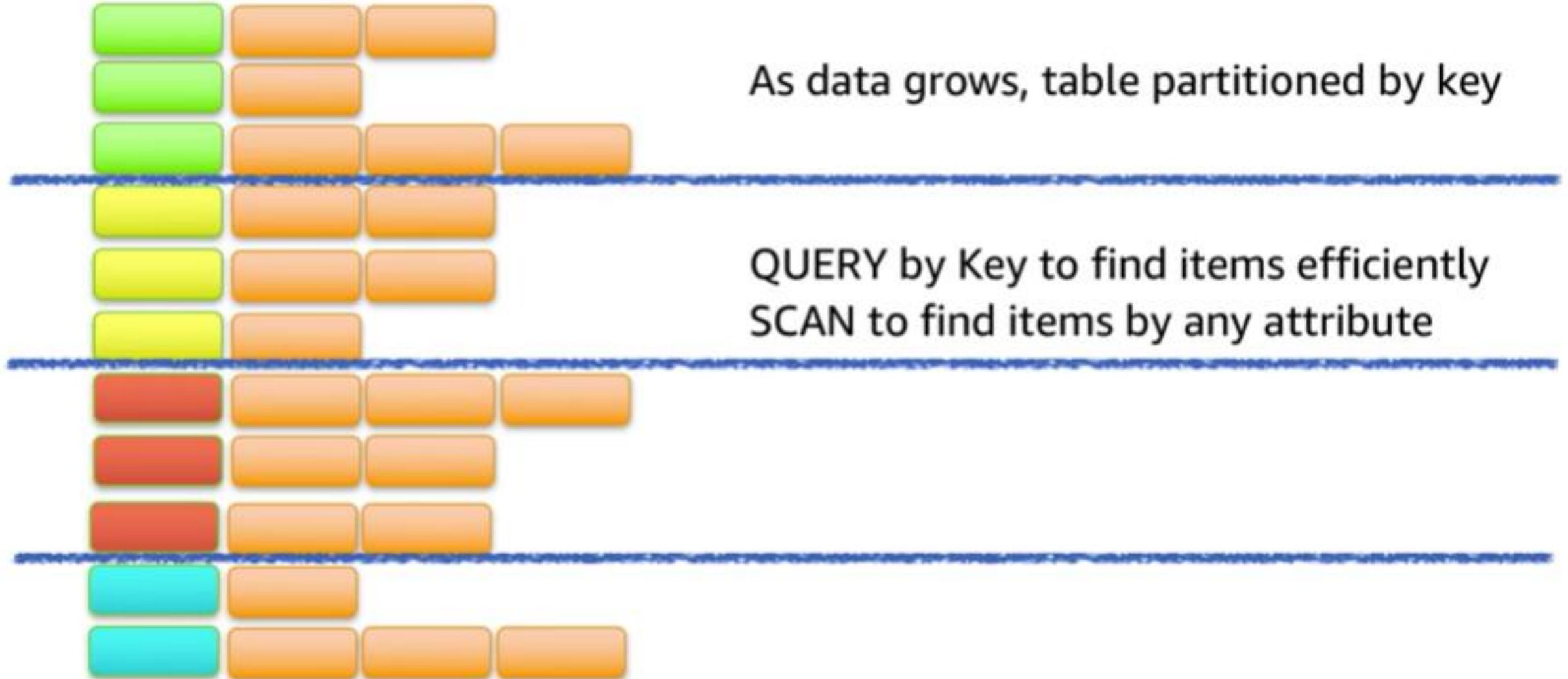
- NoSQL database tables
- Virtually unlimited storage
- Items can have differing attributes
- Low-latency queries
- Scalable read/write throughput

Amazon DynamoDB core components



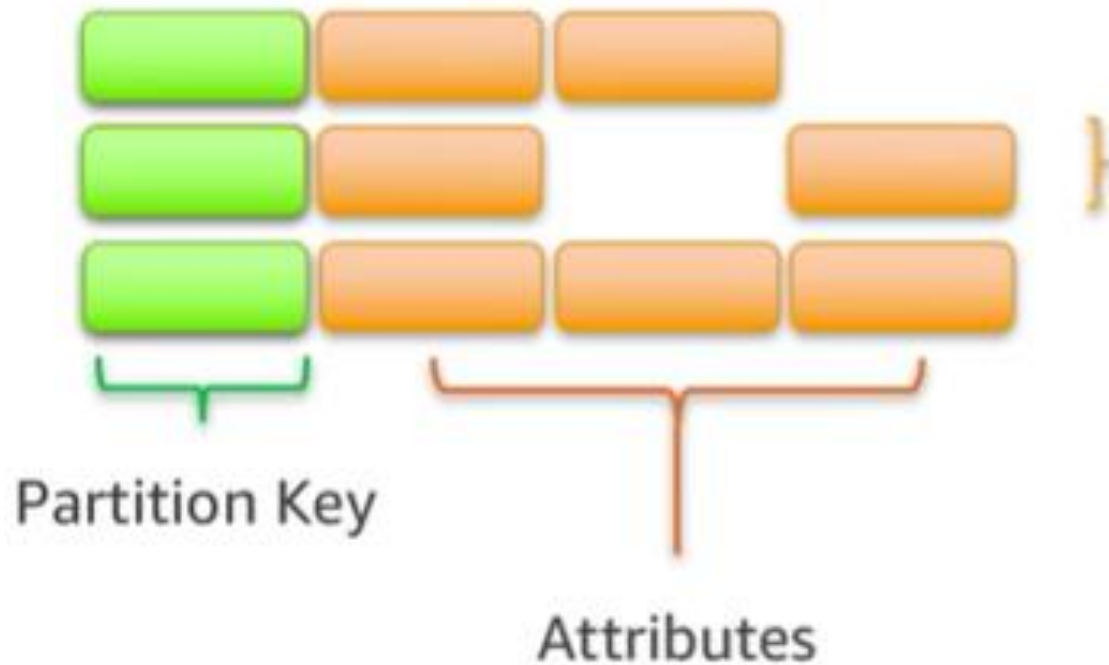
- Tables, items, and attributes are the core DynamoDB components
- DynamoDB supports two different kinds of primary keys: Partition key and partition and sort key

Partitioning

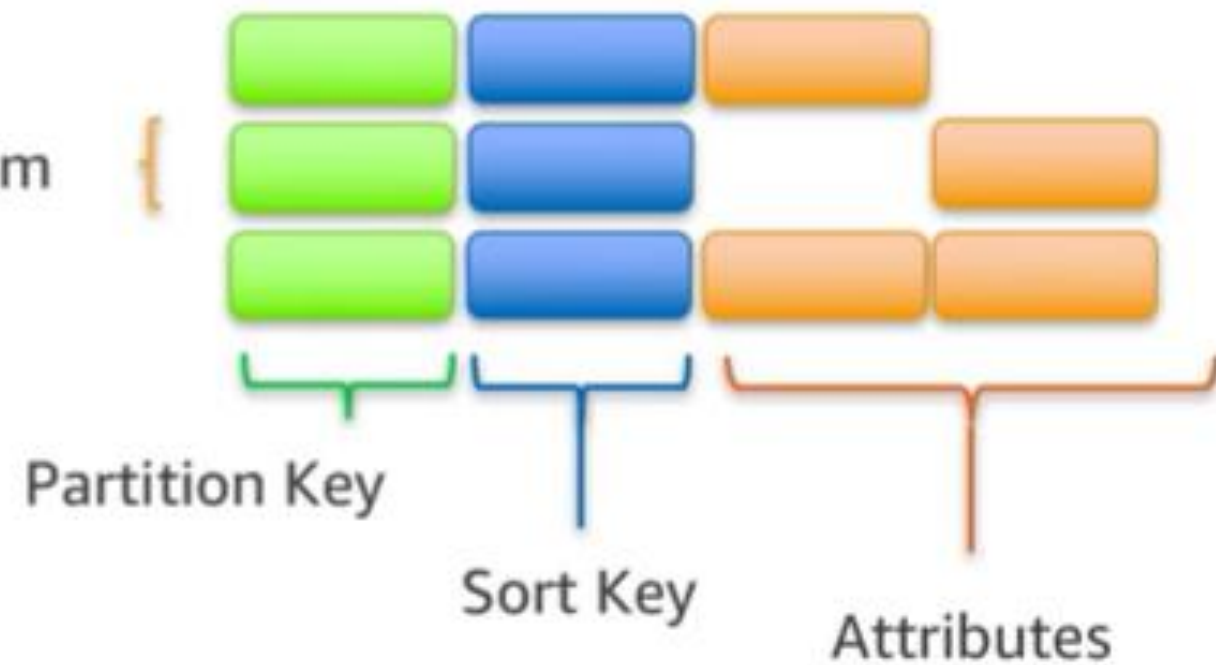


Items in a table must have a key

Single Key



Compound Key



Section 2 key takeaways



Amazon DynamoDB:

- Runs exclusively on SSDs.
- Supports document and key-value store models.
- Replicates your tables automatically across your choice of AWS Regions.
- Works well for mobile, web, gaming, adtech, and Internet of Things (IoT) applications.
- Is accessible via the console, the AWS CLI, and API calls.
- Provides consistent, single-digit millisecond latency at any scale.
- Has no limits on table size or throughput.

Recorded demo: Amazon DynamoDB



Set up demo

Amazon DynamoDB

Amazon DynamoDB demonstration



Amazon DynamoDB

Amazon DynamoDB is a fast and flexible NoSQL database service for all applications that need consistent, single-digit millisecond latency at any scale. Its flexible data model and reliable performance make it a great fit for mobile, web, gaming, ad-tech, IoT, and many other applications.

Create table

[Getting started guide](#)



Create tables



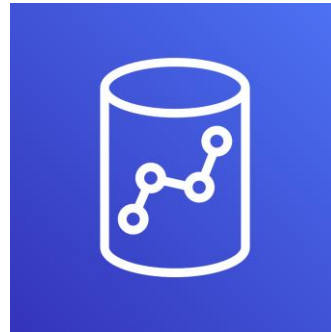
Add and query items



Monitor and manage tables

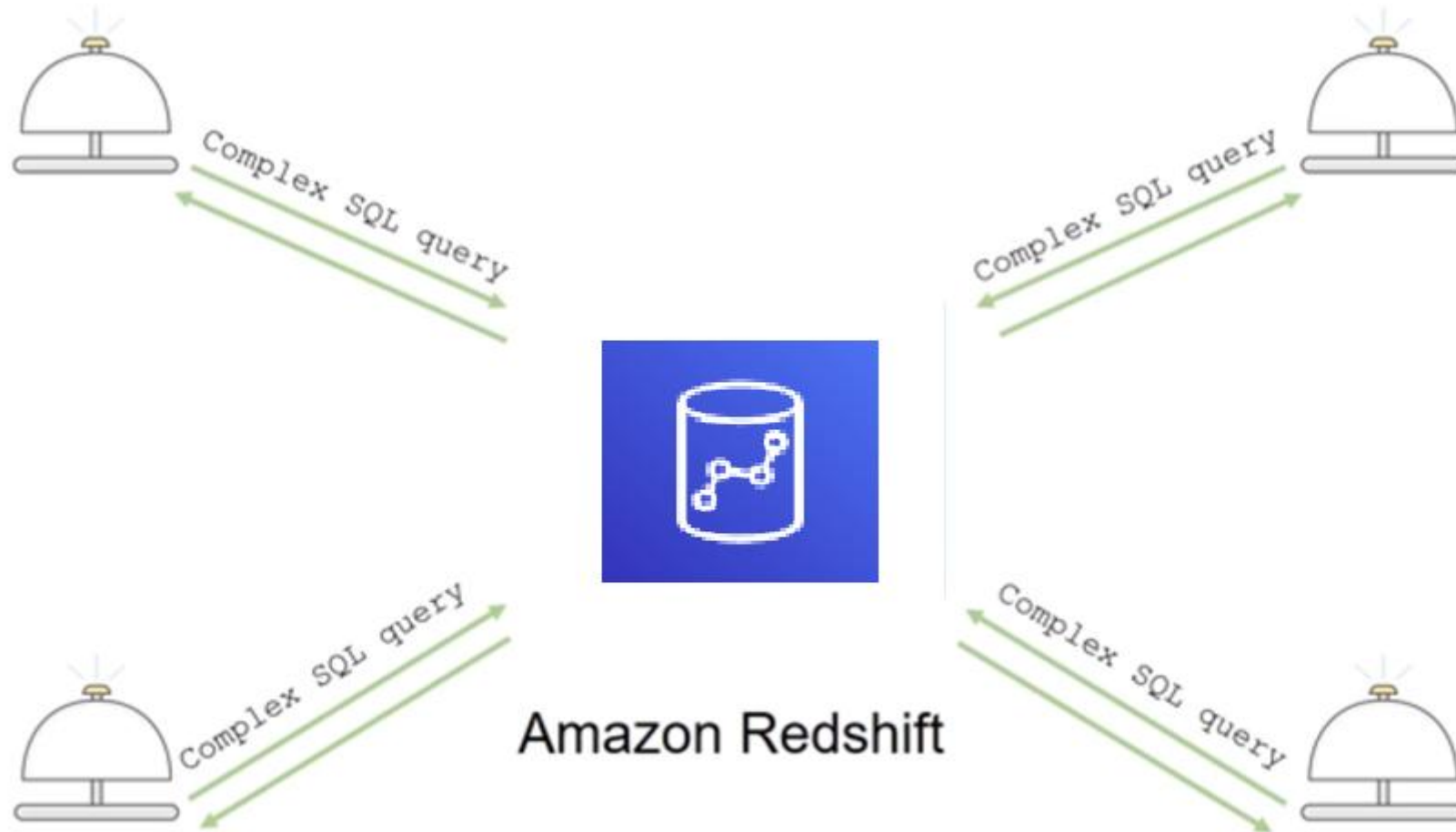
Module 8: Databases

Section 3: Amazon Redshift

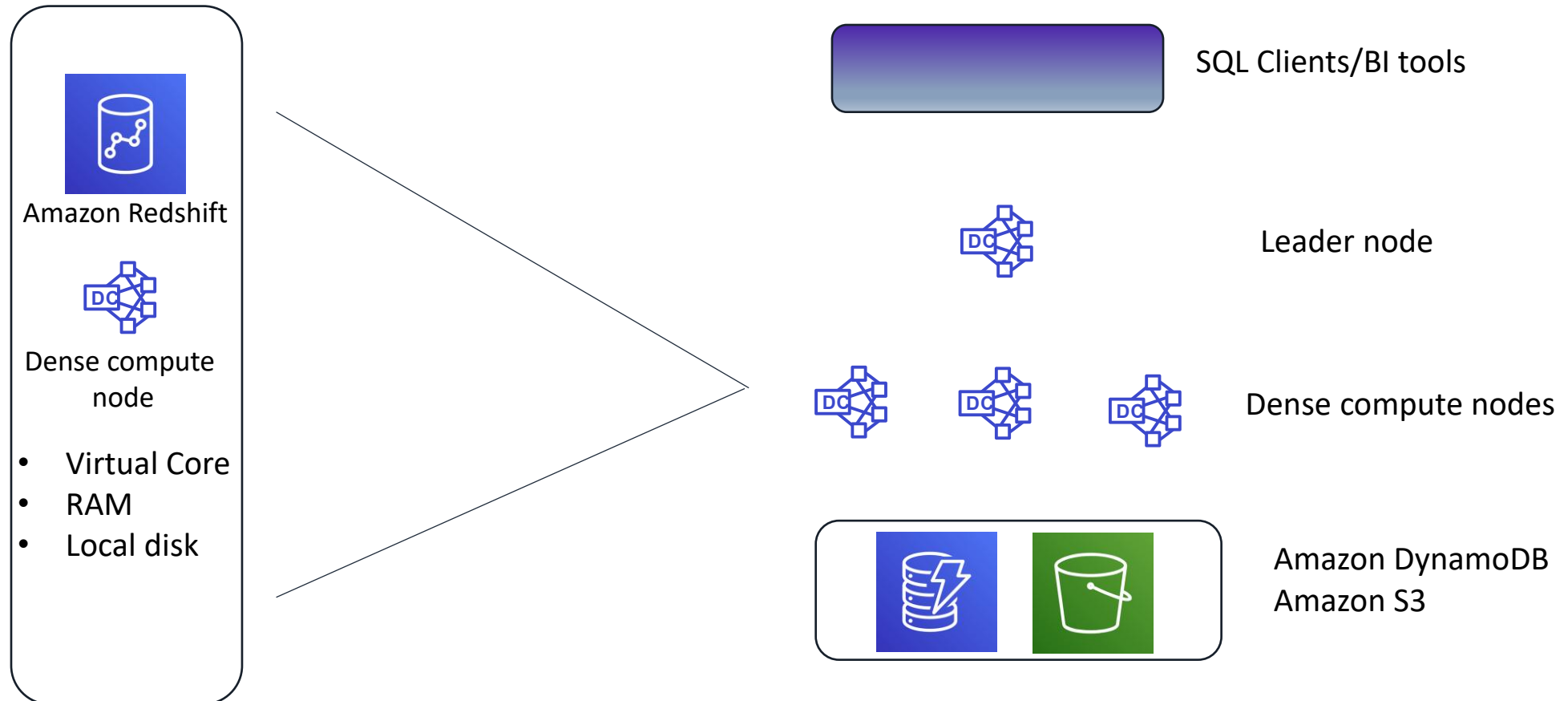


Amazon Redshift

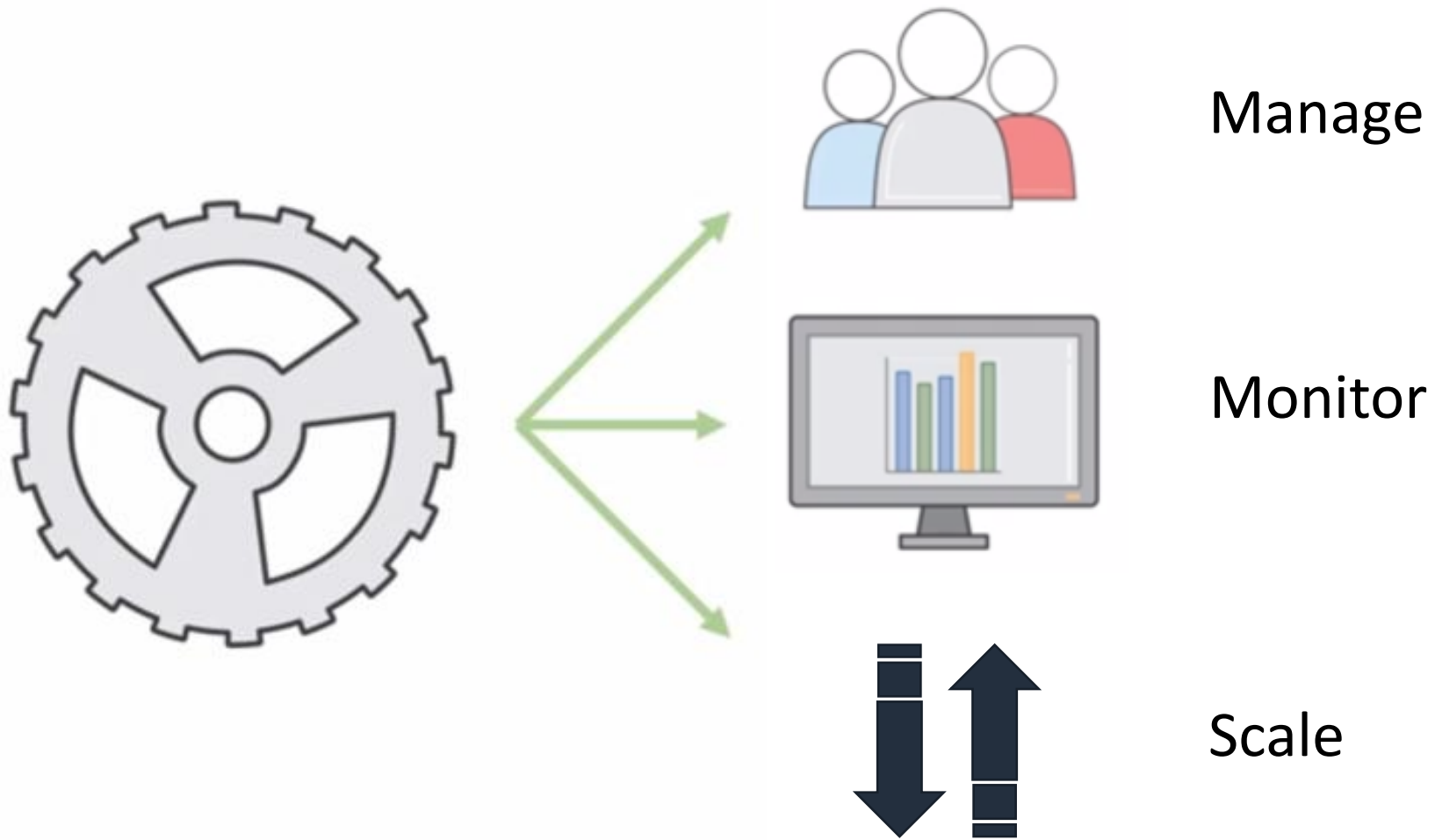
Introduction to Amazon Redshift



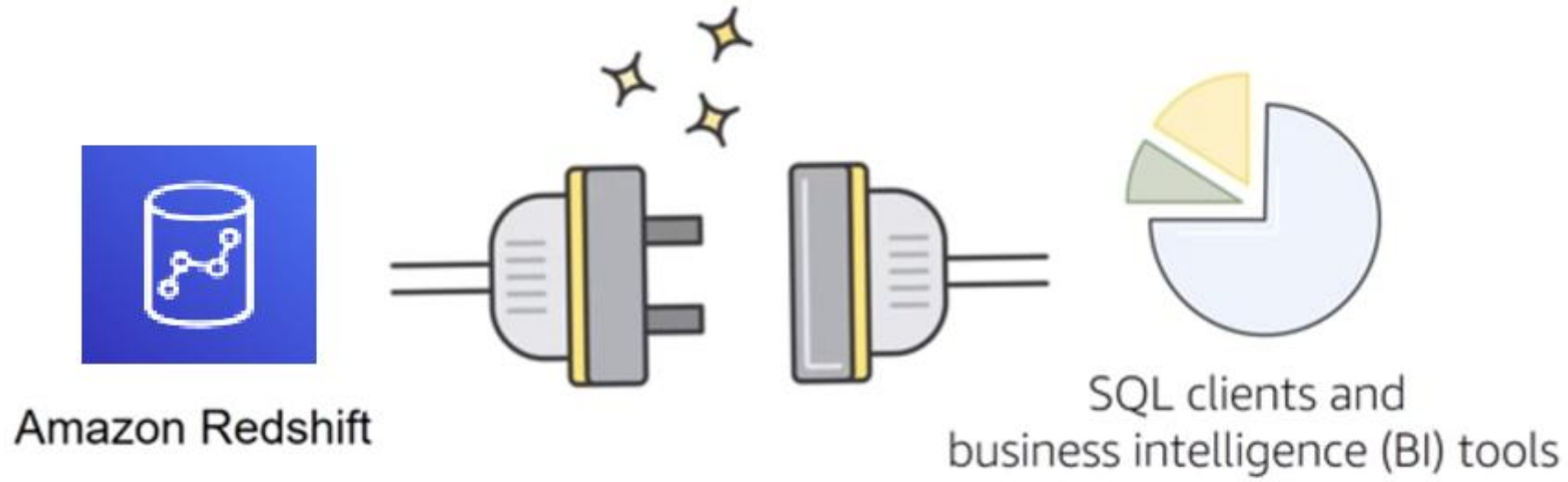
Parallel processing architecture



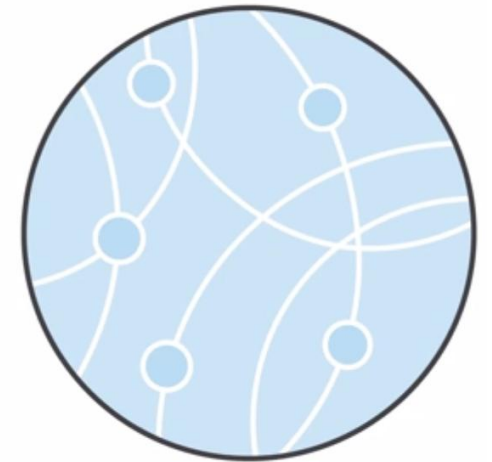
Automation and scaling



Compatibility



- Enterprise data warehouse (EDW)
 - Migrate at a pace that customers are comfortable with
 - Experiment without large upfront cost or commitment
 - Respond faster to business needs
- Big data
 - Low price point for small customers
 - Managed service for ease of deployment and maintenance
 - Focus more on data and less on database management



Amazon Redshift use cases 2

- Software as a service (SaaS)
 - Scale the data warehouse capacity as demand grows
 - Add analytic functionality to applications
 - Reduce hardware and software costs



Section 3 key takeaways

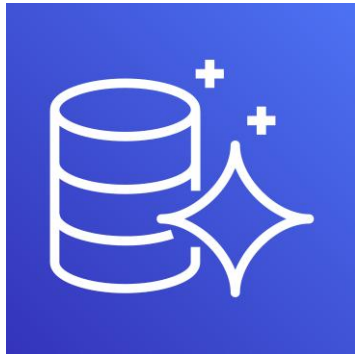


Amazon Redshift features:

- Fast, fully managed data warehouse service
- Easily scale with no downtime
- Columnar storage and parallel processing architectures
- Automatically and continuously monitors cluster
- Encryption is built in

Module 8: Databases

Section 4: Amazon Aurora



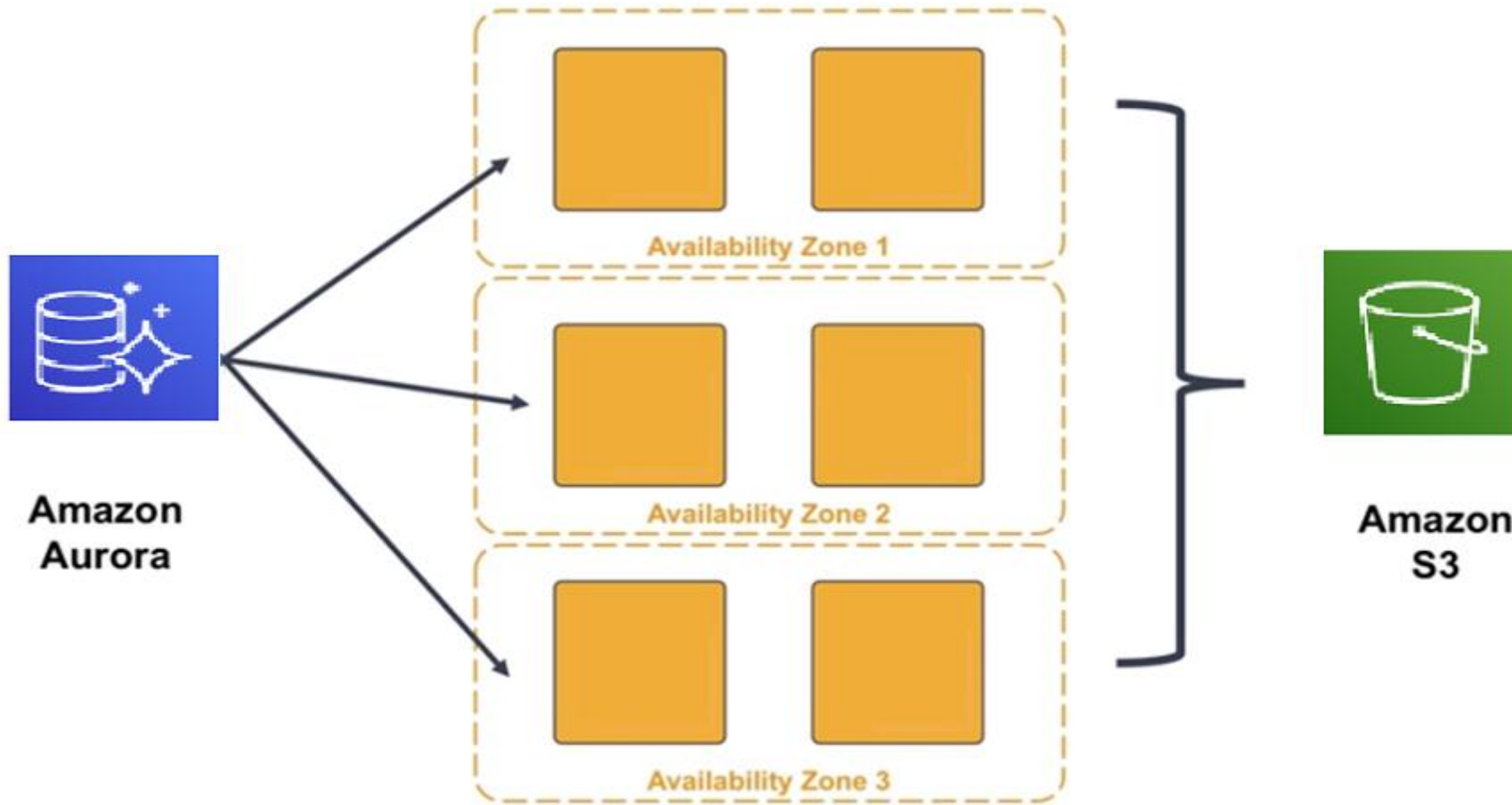
Amazon Aurora

- Enterprise-class relational database
- Compatible with MySQL or PostgreSQL
- Automate time-consuming tasks (such as provisioning, patching, backup, recovery, failure detection, and repair).

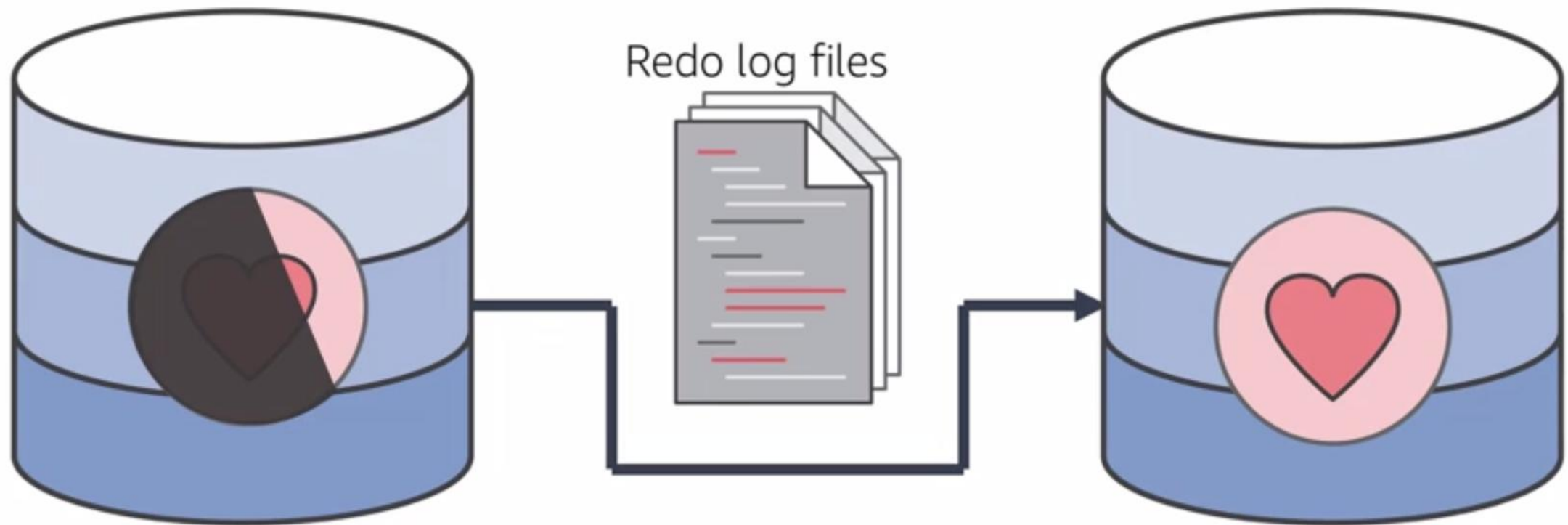
Amazon Aurora service benefits



High availability



Resilient design



Section 4 key takeaways



Amazon Aurora features:

- High performance and scalability
- High availability and durability
- Multiple levels of security
- Compatible with MySQL and PostgreSQL
- Fully managed

The right tool for the right job

What are my requirements?

Enterprise-class relational database

Amazon RDS

Fast and flexible NoSQL database service for any scale

Amazon DynamoDB

Operating system access or application features that are not supported by AWS database services

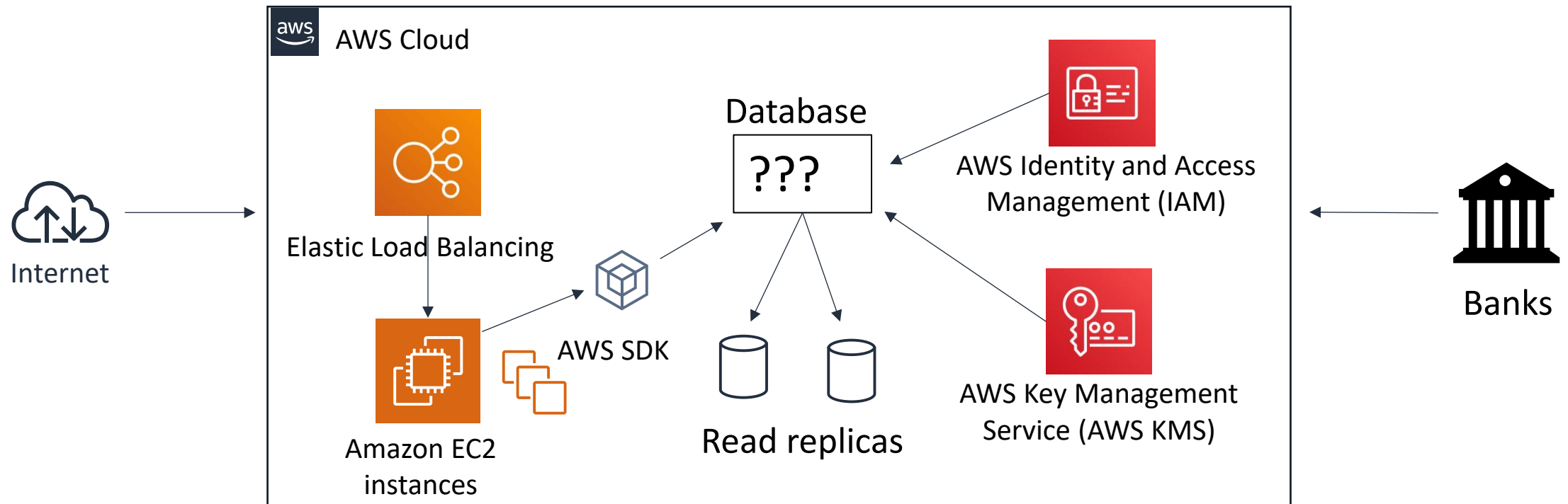
Databases on Amazon EC2

Specific case-driven requirements (machine learning, data warehouse, graphs)

AWS purpose-built database services

Database case study activity

Case : An online payment processing company that processes over 1 million transactions per day. They must provide services to ecommerce customers who offer flash sales (sales that offer greatly reduced prices for a limited time), where demand can increase by 30 times in a short time period. They use IAM and AWS KMS to authenticate transactions with financial institutions. They need high throughput for these peak loads. The following diagram illustrates their architecture.



Module 8: Databases

Module wrap-up

In summary, in this module, you learned how to:

- Explain Amazon Relational Database Service (Amazon RDS)
- Identify the functionality in Amazon RDS
- Explain Amazon DynamoDB
- Identify the functionality in Amazon DynamoDB
- Explain Amazon Redshift
- Explain Amazon Aurora
- Perform tasks in an RDS database, such as launching, configuring, and interacting

Complete the knowledge check



Sample exam question

Which of the following is a fully-managed NoSQL database service?

- A. Amazon Relational Database Service (Amazon RDS)
- B. Amazon DynamoDB
- C. Amazon Aurora
- D. Amazon Redshift

Additional resources

- [AWS Database page](#)
- [Amazon RDS page](#)
- [Overview of Amazon database services](#)
- [Getting started with AWS databases](#)

Thank you

© 2019 Amazon Web Services, Inc. or its affiliates. All rights reserved. This work may not be reproduced or redistributed, in whole or in part, without prior written permission from Amazon Web Services, Inc. Commercial copying, lending, or selling is prohibited. Corrections or feedback on the course, please email us at: aws-course-feedback@amazon.com. For all other questions, contact us at: <https://aws.amazon.com/contact-us/aws-training/>. All trademarks are the property of their owners.

